Solvency Field Theory: An Information-Theoretic Foundation for Cosmology and Gravity

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Abstract

The universe has a history because information processing is irreversible; expansion is the shape of that history at cosmological scale, and gravity is its shape at every other scale. This is the central claim of Solvency Field Theory (SFT). The framework rests on four axioms drawn from established physics — Landauer's principle, the Connes-Rovelli thermal time hypothesis, the holographic bound, and the identification of cosmic expansion as the geometric cost of maintaining an irreversible information ledger — and the master equation follows as a thermodynamic identity within the stated axiomatic scaffold rather than as a postulated dynamical law. The physical mechanism implementing the axioms is the Compton Maintenance Postulate: every massive particle maintains its quantum existence through continuous phase evolution at the Compton frequency , with each complete phase cycle constituting one irreversible bit of information and demanding one Planck area of cosmological boundary surface to accommodate it.

From this foundation we derive a modified Friedmann equation , differing from the standard result by a content-independent factor of exactly , and a galactic acceleration threshold m/s confirmed against 137 SPARC galaxies at 63% error reduction with zero free parameters, with morphology-dependent confirmed at . The factor is not a single unexplained constant: it appears in five distinct physical contexts, in each case decomposing into the same two independent factors of — one from the phase convention (versus) and one from the boundary-versus-bulk normalization (Bekenstein-Hawking versus Einstein coupling). Four of these appearances are derived; the fifth, a candidate splice in the cosmic microwave background transfer coefficient, is presented at candidate-derivation status with its physical motivation explicit.

The framework's historical cosmology pipeline produces a dark energy equation of state (,) matching DESI DR2 at a CPL-plane distance of 0.005, with phantom crossing at and CMB compatibility at 0.078% RMS against Planck 2018 across multipoles 2–2500. A dimensionally corrected reconstruction of from first principles has been derived and is presented alongside the historical pipeline; full reconciliation of the two through a complete rerun against the original channel history is flagged as an audit item for a V10.1 update. The framework recovers the exact Schwarzschild exterior metric and the exact Tolman-Oppenheimer-Volkoff

hydrostatic equations in the strong-field limit as the leading closure of the local information-maintenance balance. A single thermodynamic mechanism — forced transitions of the maintenance process when its current regime can no longer be sustained — structurally unifies the early-universe cosmic microwave background imprint with the late-universe dark energy phantom crossing, producing observable scars at both ends of cosmic history. Falsifiable predictions include environment-dependent structure growth testable by DESI DR3 and Euclid, specific growth-rate splits between voids and overdensities, and a candidate CMB transfer signature. Open research fronts include the formal unification of the early and late-universe forced transitions, a first-principles derivation of the boundary-bulk normalization factor, and closure of the candidate CMB transfer splice.

1. Introduction: An Information-Theoretic Foundation

The universe has a history because information processing is irreversible; expansion is the shape of that history at cosmological scale, and gravity is its shape at every other scale. This sentence is the thesis of Solvency Field Theory, and the paper that follows is a sustained argument for it.

Conventional theoretical physics treats general relativity as fundamental and cosmology as its large-scale application. Solvency Field Theory proposes the opposite ordering. The universe's history — the continuous, irreversible accumulation of processed information, paid for thermodynamically and accommodated geometrically — is the primary object. Cosmological expansion is what that history looks like at the largest scale, where the maintenance cost is averaged over horizon volumes. Gravity is what the same history looks like at every smaller scale, where the maintenance cost is resolved into local curvature. The two are not independent phenomena governed by a single theory of dynamics; they are two geometric shapes of one thermodynamic process.

This is a specific commitment about what kind of object the universe is, and it is the commitment that the rest of the paper defends. SFT is not a modification of general relativity. It introduces no new fields, no new particles, no new dimensions, and no adjustable parameters beyond those already present in standard physics. It is not a phenomenological fit to cosmological data, nor a reformulation of gravity in new variables, nor a proposal for quantum gravity. It is an information-theoretic foundation: a claim that general relativity, the Friedmann equations, and the galactic acceleration scale are all consequences of a single thermodynamic identity that follows from four axioms drawn from established physics.

The four axioms are: (1) information processing has an irreducible thermodynamic cost (Landauer 1961); (2) time is the macroscopic perception of irreversible state changes (Connes-Rovelli thermal time hypothesis, 1994); (3) the holographic bound on information density is absolute (Bekenstein 1973, 't Hooft 1993, Susskind 1995); and (4) cosmic expansion is the geometric response required to accommodate the growing information budget without violating the first three. From these axioms, the master equation emerges as the native thermodynamic identity of the framework within the stated scaffold. Section 3 develops this foundation in detail. Section 4 introduces the Compton Maintenance Postulate as the physical mechanism by which the axioms are implemented: every massive particle maintains its

quantum existence by continuously processing information at its Compton frequency, and each complete phase cycle is charged one Planck area on the cosmological boundary.

The paper's structural argument has three load-bearing claims. First, that a single geometric identity — — appears in five distinct physical contexts, in each case decomposing into the same two independent factors of , and this structural recurrence is the signature of a unified underlying mechanism rather than a sequence of coincidences. Second, that the maintenance process undergoes forced thermodynamic transitions when its current regime can no longer be sustained, and that two such transitions — one at the early universe, one at the late universe — leave observable scars on the cosmic microwave background and the dark energy equation of state respectively. Third, that general relativity itself is the strong-field limit of the local information-maintenance balance, with the exact Schwarzschild metric and the exact Tolman-Oppenheimer-Volkoff equations emerging as the leading forced solutions of the local balance rather than as postulated dynamics.

The quantitative results supporting these claims are: a modified Friedmann equation with zero free parameters, confirmed against the standard equation by the structural ratio; a galactic acceleration threshold confirmed against 137 SPARC galaxies at 63% error reduction with a morphology-dependent parameter confirmed at ; a dark energy equation of state matching DESI DR2 at 0.005 in the CPL plane through the historical cosmology pipeline, with a dimensionally corrected reconstruction of the underlying chain flagged for V10.1 audit; CMB compatibility at 0.078% residual against Planck 2018; and recovery of exact general relativity in the strong-field limit. Sections 5 through 12 present each of these results with enough mathematical detail for a reader to follow the logical skeleton of the derivation; the full derivations are in the companion document.

The paper is organized as follows. Section 2 states the three unsolved problems in modern cosmology that SFT addresses as manifestations of a single geometric identity. Section 3 establishes the four axioms and derives the master equation. Section 4 introduces the Compton Maintenance Postulate. Section 5 derives the modified Friedmann equation. Section 6 presents the galactic confirmation via SPARC. Section 7 develops the five-appearance structural unification of . Section 8 promotes the master equation to a variational principle via the holographic count equation and the constrained action. Section 9 establishes the general principle of forced thermodynamic transitions of the maintenance process. Sections 10 and 11 deliver the late-universe and early-universe instances of this principle in quantitative detail. Section 12 presents the strong-field closure to exact general relativity. Section 13 lists the falsifiable predictions. Section 14 states the open theorems and research targets. Section 15 concludes.

2. Three Unsolved Problems

Three puzzles in modern cosmology share a common feature: each involves an unexplained numerical scale that connects quantum mechanics to the large-scale structure of the universe. Despite decades of theoretical effort, none has a consensus physical explanation. This section describes them. The remainder of the paper argues that all three are manifestations of a single geometric identity, and that this identity emerges from a single thermodynamic mechanism.

The dark energy problem. The observed accelerated expansion of the universe requires an energy density parameterized as in the standard CDM model. This number has no theoretical derivation. The naïve quantum field theory estimate of vacuum energy density, obtained by summing zero-point contributions up to the Planck cutoff, exceeds the observed value by approximately — the largest discrepancy between theory and observation in the history of physics. The standard approach treats as a free parameter fitted to observational data. Recent DESI DR2 measurements have sharpened the situation further: the dark energy equation of state appears to evolve with redshift, with and , ruling out a static cosmological constant at 3.4–5.4 depending on dataset combination. There is no accepted physical mechanism that produces this specific equation-of-state evolution from first principles.

The MOND acceleration scale. Galaxy rotation curves deviate from Newtonian predictions below a characteristic acceleration threshold m/s, identified empirically by Milgrom in 1983 and confirmed across hundreds of galaxies over four decades. The standard cold dark matter approach fits these rotation curves with tunable halo parameters, but it does not explain why a single universal acceleration scale governs the transition for all galaxies regardless of mass, morphology, or environment. The approximate numerical coincidence between and has been noted repeatedly in the literature, including by Milgrom himself, but has never been derived from a physical mechanism. It has been treated as either coincidental or suggestive, and its unexplained status has persisted for forty years.

The holographic capacity gap. The Bekenstein-Hawking entropy of the cosmological horizon, computed by applying the area law to a Hubble-sized surface, is approximately bits. The total entropy produced by all astrophysical processes over the entire history of the universe — stellar fusion, black hole formation, gravitational collapse, nucleosynthesis — amounts to approximately cumulative operations. The ratio between the boundary's information capacity and the bulk's processing history exceeds . No standard framework explains why the cosmological boundary appears so vastly over-provisioned relative to the physical processes it contains. The gap is acknowledged in the literature but not addressed: it is simply the quantitative expression of how little of the holographic bound the universe actually uses.

These three problems are conventionally treated as independent. The dark energy problem is an issue of vacuum energy and cosmological fine-tuning. The MOND scale is an issue of galactic dynamics and modified gravity. The holographic capacity gap is an issue of quantum gravity and information theory. They appear in different subfields, involve different empirical regimes, and are pursued by different research communities.

This paper argues that all three are manifestations of a single geometric identity: . The identity arises from two independent unit conversions at the interface between quantum phase and cosmological geometry, it decomposes into the same two factors of in each context, and it appears whenever the Planck scale (defined using the reduced Planck constant) interfaces with the Compton scale (defined using the linear Planck constant) within a cosmological boundary. Section 7 develops this unification in detail. Sections 5, 6, and 10 show that the three problems' quantitative signatures — the modified Friedmann equation, the MOND acceleration scale, the holographic capacity gap, and now also the DESI dark energy equation of state — all fall out of the same underlying information-maintenance mechanism. The unification is not an interpretation layered on top of existing results; it is the direct algebraic consequence of the four axioms developed in the next section.

3. The Four Axioms of Solvency Field Theory

Solvency Field Theory rests on four axioms. Each is drawn from an established result in twentieth- or early twenty-first-century physics. None is introduced by SFT itself; the framework's contribution is to commit to all four simultaneously and to derive their joint consequences. This section states the axioms, explains what physical commitments they encode, and demonstrates that the master equation follows as the native thermodynamic identity of the framework from the four axioms taken together.

Axiom 1 — Information has thermodynamic cost. *Landauer's principle (Landauer 1961).* The erasure of one bit of information dissipates at least of energy as heat, where is Boltzmann's constant and is the temperature of the environment absorbing the dissipated heat. This is not a statement about any specific physical system; it is a general consequence of the second law of thermodynamics applied to computational processes. It has been experimentally confirmed in single-molecule, single-electron, and single-spin systems, and it has never been violated in any experimental regime. For SFT, Axiom 1 establishes that irreversible information processing cannot be thermodynamically free: every bit has a cost, and that cost must be paid somewhere in the physical system that performs the processing. Structure formation, nucleosynthesis, stellar radiation, gravitational collapse, and black hole accretion are not exceptions to this principle — they are specific physical manifestations of it at astrophysical scales. The universe is, in this sense, a computer: not in the metaphorical sense sometimes encountered in popular accounts, but in the specific technical sense that its irreversible dynamics are subject to Landauer's bound.

Axiom 2 — Time is the entropy clock. *The Connes-Rovelli thermal time hypothesis (Connes and Rovelli 1994).* Time is not a background parameter that the universe is embedded in. It is the macroscopic perception of irreversible state changes in the physical degrees of freedom that constitute the universe. A system that ceases to produce entropy has no internal clock; there is nothing to count. Conversely, a system whose entropy production rate determines its dynamical evolution rate is one in which time is identical to the accumulation of irreversible processing. The thermal time hypothesis was introduced by Connes and Rovelli to address the problem of time in quantum gravity, where the usual spacetime notion of time becomes ill-defined. For SFT, Axiom 2 is load-bearing in a specific way: it is what makes the identification structurally natural rather than arbitrary. If time itself is the accumulation of irreversible bit-writes to the universe's ledger, then the rate of cosmic expansion — which is a rate measured per unit time — is necessarily tied to the fractional rate at which the ledger is being filled. The master equation is not an additional postulate layered on top of the axioms; it is what the axioms are about.

Axiom 3 — The holographic bound is absolute. *The Bekenstein bound and the holographic principle (Bekenstein 1973, Hawking 1975, 't Hooft 1993, Susskind 1995).* The maximum information content of a bounded spatial region scales with the region's boundary area, not its volume, and this bound is saturated on gravitational horizons. Explicitly, a horizon of area has information capacity bits, where is the Planck area. This result was originally derived for black hole event horizons and subsequently generalized to cosmological horizons via apparent-horizon thermodynamics (Jacobson 1995, Padmanabhan 2010). For SFT, Axiom 3 establishes that the universe's ability to store information is finite and geometrically constrained: the

cosmological boundary can accommodate a specific number of Planck-area cells, and no more. Any information-producing process operating in the bulk must ultimately be accommodated by this boundary, and the boundary's capacity is set by the horizon's geometry.

Axiom 4 — Expansion is the geometric cost of maintenance. The previous three axioms, taken jointly, imply the fourth: as the universe generates irreversible information (Axiom 1), and as time is the accumulation of that generation (Axiom 2), and as the boundary has finite capacity (Axiom 3), the cosmological geometry must expand to accommodate the growing information budget. Expansion is not a force pushing outward, and it is not driven by a vacuum energy or a cosmological constant. It is the geometric response consistent with the first three axioms: the way for the universe to continue processing information without violating the holographic bound is for the boundary's capacity to grow in step with the bulk's processing. Axiom 4 is sometimes described as "the universe pays its observation debt through geometry," and this language is accurate but should not be read metaphorically. The debt is thermodynamically real (Axiom 1), it accumulates in proper time (Axiom 2), and it is capped by holographic capacity (Axiom 3). Expansion is what the universe does to stay solvent against its own computation.

A preemptive clarification. A careful reader may observe that Axiom 4 appears to encode part of the master equation's conclusion in its statement. The response is that Axiom 4 commits to the *kind* of geometric response — expansion accommodates the information budget — without specifying the *rate*. The specific fractional rate emerges only when one requires that the geometric accommodation proceed at the pace set by the bulk processing itself, which is a further consistency condition between Axioms 1 and 3 under the Axiom 2 identification of time with irreversibility. The master equation is therefore not circular relative to the axioms; it is the unique fractional rate consistent with all four within the stated scaffold. A reader who rejects any of the four axioms is free to do so, but the joint commitment is what SFT stands on.

With the four axioms in place, the master equation follows. The argument is a native identity claim within the scaffold, and we present it in three-sentence form.

Starting point. Let denote the total rate of irreversible information production in the universe (Axiom 1, made quantitative by Axiom 2), and let denote the current information capacity of the cosmological boundary (Axiom 3). Both quantities are computable from established physics and have definite values at each moment of cosmic time.

Pivot. The cosmological geometry must expand at a rate such that the boundary's capacity grows in step with the bulk's processing . The fractional rate at which the boundary's capacity is being filled by bulk processing is . The fractional rate at which the cosmological geometry is expanding is, by definition, the Hubble parameter . By Axiom 4, these two rates must be equal — any difference would either violate the holographic bound (if) or leave boundary capacity unused while the bulk continues processing (if , which would mean the universe is expanding faster than required to accommodate its computation, implying a free parameter or an external driver that the axioms explicitly rule out).

Result. The Hubble parameter is therefore fixed by the identity

This is the master equation of Solvency Field Theory. It is not a postulated dynamical law. It is the native thermodynamic identity of the framework, derived within the stated axiomatic scaffold in the same sense that the ideal gas law is a thermodynamic identity for non-interacting point particles: it is the definition of what temperature *is* for that system, and here is the definition of what the Hubble rate *is* for a holographically bounded irreversible-computation universe.

A brief clarification of what the master equation does and does not claim. It does not claim that the universe is literally a classical computer, nor that the boundary is a physical memory device, nor that and are fundamental quantities more basic than spacetime. These would be additional metaphysical commitments that the axioms do not make. What the master equation claims is narrower: given that information processing is irreversible (Axiom 1), that time is the accumulation of that processing (Axiom 2), and that the boundary has finite capacity (Axiom 3), the cosmological expansion rate is algebraically fixed by the ratio of the two rates. Any theory that accepts the four axioms must also accept as a consequence. SFT's distinctive claim is not the master equation itself but the commitment to all four axioms simultaneously.

The master equation is the foundation from which the rest of the paper is built. Section 4 introduces the physical mechanism that makes and calculable: the Compton Maintenance Postulate. Section 5 applies the master equation to derive the modified Friedmann equation. Sections 6 through 12 develop the consequences at galactic, structural, cosmological, and strong-field scales. Every quantitative result in the paper traces back to and the four axioms that ground it.

4. The Compton Maintenance Postulate: The Mechanism

Section 3 grounded the master equation as the native thermodynamic identity following from the four axioms. For the equation to produce quantitative predictions, the total information production rate and the boundary capacity must be calculable from known physics. This section introduces the physical mechanism that makes them calculable: the Compton Maintenance Postulate.

The postulate is the physical mechanism by which SFT's axioms are implemented — it tells us what the universe's irreversible bit-writes physically *are*. The axioms establish that information processing has cost, that time is the accumulation of that processing, and that the boundary has finite capacity. The Compton Maintenance Postulate identifies the specific physical process that counts as one bit and the specific geometric cost that counts as one Planck-area debit.

The physical fact. A massive particle is not a static object in quantum mechanics. Its quantum state evolves continuously in time, and the rate of this evolution is set by the particle's rest energy through the de Broglie relation. Specifically, the quantum phase of a massive particle rotates at the Compton frequency:

where is the particle's rest mass, is the speed of light, and is Planck's constant. This is not a postulate of SFT; it is a measurable property of any massive quantum state, established by

Compton scattering experiments, matter-wave interferometry, and the de Broglie relation itself. A particle that stops evolving its phase at the Compton frequency ceases to exist as a massive quantum object. Phase evolution at is the minimal dynamical activity required for a particle to be what it is.

The postulate. Solvency Field Theory proposes the following physical identification:

The Compton Maintenance Postulate. Every massive particle maintains its quantum existence through continuous phase evolution at the Compton frequency. Each complete phase cycle — radians of quantum phase, one linear period of — constitutes one irreversible bit of information, and demands exactly one Planck area () of cosmological boundary surface to accommodate it.

Three distinct physical claims are embedded in this postulate, and each must be evaluated on its own merits.

Claim 1: Existence is computational. A massive particle persists by continuously performing one irreversible information operation per Compton cycle. This is not a metaphor. Phase evolution at the Compton frequency is a physical process with a definite rate, and each complete cycle represents an irreversible transition in the particle's quantum history — the state at the end of the cycle cannot be reversed to the state at the beginning without thermodynamic cost. This identification is consistent with Axiom 1 (Landauer): the continuous phase evolution is the specific physical manifestation of the universe performing irreversible computation, and the Compton frequency is the rate at which that computation proceeds for each massive particle.

Claim 2: One bit equals one complete cycle, not one radian of phase advance. The fundamental unit of irreversibility is the full revolution of quantum phase, corresponding to the linear Compton frequency , not the angular Compton frequency . This is a specific testable choice that makes a sharp numerical prediction: all SFT formulas involving mass-energy processing use rather than , and every appearance of in a final result traces to this choice. If one bit corresponded to one radian of phase advance instead of one complete cycle, the theory's numerical predictions would differ by factors of in specific places — places that Section 6's SPARC data can directly discriminate. The SPARC test will show that the data prefer the complete-cycle convention at a level that rules out the one-radian alternative.

Claim 3: The geometric cost is one Planck area per bit. Each irreversible bit demands one Planck area m of cosmological boundary surface. This identification is consistent with Axiom 3 (the holographic bound), which establishes that a boundary of area has capacity bits. The Compton Maintenance Postulate promotes this from a statement about the maximum capacity of a boundary — a state equation — to a statement about the continuous accommodation of bulk processing by boundary growth — a process equation. Information is being written to the boundary, one Planck area at a time, by the continuous phase maintenance of every massive particle in the universe. The factor of 4 in the Bekenstein-Hawking formula (which SFT retains) reflects the geometric convention for counting horizon degrees of freedom and will become load-bearing in Section 7 when the two-factor decomposition of is developed.

With the postulate in place, the two quantities needed to evaluate the master equation are calculable. The total information production rate is the sum of Compton frequencies over all

massive particles in the universe. For a universe with total mass-energy enclosed within the Hubble volume:

This expression counts the number of complete phase cycles performed by all massive particles per unit time. Its units are bits per second, and its value is set entirely by the total mass-energy content and the Planck constant. Note specifically that appears rather than , reflecting Claim 2 above: the postulate identifies one bit with one complete cycle. This is the choice that will produce a factor of in the final result and that SPARC will test directly.

The boundary capacity is given by the Bekenstein-Hawking formula applied to the Hubble horizon. The Hubble horizon is the sphere of radius centered on the observer, and it has area . Its information capacity is:

This is the total number of Planck-area cells on the Hubble horizon at a given moment. Its units are bits, and its value depends on and on the Planck area. The choice of the Hubble horizon as the relevant boundary is not arbitrary: in flat FLRW spacetime, the Hubble sphere coincides with the apparent horizon, which is the surface that appears in rigorous cosmological thermodynamic derivations (Jacobson 1995, Padmanabhan 2010) and is the unique causal boundary within which information can be exchanged in one Hubble time. The companion document presents the full apparent-horizon thermodynamics derivation; for the main paper, it suffices to note that the Hubble horizon is forced by causal consistency, not chosen by convenience.

With and both calculable, the master equation becomes a closed algebraic relation between the Hubble parameter and the matter content of the universe. Section 5 carries out the substitution and derives the resulting modified Friedmann equation.

5. The Modified Friedmann Equation

Section 4 provided the two ingredients needed to evaluate the master equation: the total information production rate and the Hubble horizon information capacity . Substituting these into the master equation from Section 3 produces a closed equation for the Hubble parameter in terms of the matter content of the universe. This section carries out the substitution explicitly, derives the resulting modified Friedmann equation, and computes its ratio to the standard Friedmann equation. The ratio turns out to be a content-independent geometric factor of exactly , which is the first appearance of the structural identity developed in Section 7.

The substitution. The master equation is:

Substituting and :

The total mass-energy enclosed within the Hubble volume is , where is the mean density and is the volume of the Hubble sphere:

The Planck area can be written in terms of the Newtonian gravitational constant and the linear Planck constant rather than the reduced Planck constant :

This substitution is the step at which the phase convention — Claim 2 of the Compton Maintenance Postulate — enters the algebra explicitly. The Planck area is conventionally defined using (angular phase), while the Compton maintenance rate is defined using (linear cycles). Rewriting in terms of introduces a factor of that will propagate through the rest of the derivation.

Substituting both expressions into the intermediate relation:

The factors of , , and one factor of cancel cleanly. What remains is:

Multiplying both sides by yields the modified Friedmann equation of Solvency Field Theory:

The result. This equation is derived from the master equation and the Compton Maintenance Postulate with no adjustable parameters, no fitting constants, and no ad hoc corrections. Given the axiomatic foundation of Section 3 and the physical mechanism of Section 4, the equation is forced. Any theory that accepts the four axioms and identifies one bit with one complete Compton cycle must produce this result.

The ratio to standard Friedmann. The standard Friedmann equation, derived from general relativity applied to an FLRW spacetime filled with pressureless dust, is:

The ratio of SFT to standard is:

The density cancels. The gravitational constant cancels. What remains is a pure geometric factor: , content-independent and universal. It does not depend on what is in the universe. It depends only on the structural difference between how SFT accounts for boundary capacity (via Bekenstein-Hawking with factor) and how standard general relativity couples matter to geometry (via the Einstein equation with coefficient), combined with the phase-convention choice of the Compton Maintenance Postulate (linear cycles via rather than angular radians via).

The appearance of this exact factor is load-bearing for the rest of the paper. Section 7 will show that is not a single unexplained constant but a geometric identity that appears in five distinct physical contexts, in each case decomposing into the same two independent factors of — one from the phase convention and one from the boundary-versus-bulk normalization. Section 6 will show that the square root of this ratio, taken at the galactic scale, produces an acceleration threshold that the SPARC data confirm at 63% error reduction with zero free parameters.

Addressing the obvious objection. A reader encountering for the first time will immediately ask: if this is the correct Friedmann equation, why does the standard work so well? The difference between the two equations is the quantitative consequence, within SFT, of making information maintenance an explicit boundary-bulk accounting constraint rather than leaving that accounting implicit. This is not a claim that standard general relativity is secretly doing SFT. It is a claim that the two frameworks handle the same underlying content through different accounting conventions, and that the specific numerical difference between them — the factor of — is structured by the same geometric identity that appears at galactic scales, in the holographic capacity gap, and in the structure of the dark energy sector.

The matter-only SFT Friedmann equation does not by itself reproduce the observed expansion rate. For , it gives , leaving a structural gap of approximately 44%. In the framework developed here, that gap is not treated as an additional free parameter but as the quantity to be explained by the maintenance-regime dynamics of Sections 9 and 10. The remainder of the paper therefore asks whether the structural identity, together with the forced-transition mechanism, can account for that gap in a way consistent with DESI, the CMB, galactic dynamics, and the strong-field limit.

6. Galactic Dynamics: The SPARC Confirmation

The modified Friedmann equation of Section 5 differs from the standard result by a content-independent factor of . Taking the square root of this ratio at the present epoch yields a natural acceleration scale for the framework, and this scale can be tested directly against galactic rotation curve data. This section carries out that test. The result is that the derived acceleration threshold, with zero free parameters, reduces the mean rotation-curve error across 137 SPARC galaxies by 63% relative to baryonic mass alone, and the data independently prefer the specific factor of that the Compton Maintenance Postulate predicts over several nearby alternatives.

The acceleration threshold. From the Friedmann ratio , taking the square root gives . Multiplying by to obtain a natural acceleration and evaluating at the present Hubble constant :

This value is derived, not fitted. It is the direct geometric consequence of the Compton Maintenance Postulate combined with the master equation, evaluated at the present cosmological epoch. The factor of is the ratio between the linear Compton frequency (which counts complete cycles) and the angular Compton frequency (which counts radians), and it enters because the Compton Maintenance Postulate identifies one irreversible bit with one complete phase cycle rather than one radian of phase advance (Claim 2 of Section 4).

MOND, as originally proposed by Milgrom in 1983, identified an empirical acceleration threshold m/s below which galactic rotation curves deviate from Newtonian predictions. The value was extracted from galactic data without any theoretical derivation, and its approximate coincidence with has been noted in the literature for four decades without being derived from a physical mechanism. The SFT value agrees with Milgrom's empirical to within 13% — and, critically, with zero free parameters. The 13% residual is addressed as a research target in Section 14; candidate explanations include QCD binding energy contributions to the effective Compton frequency and systematic uncertainties in the empirical determination of .

The SPARC test. The SPARC database (Lelli, McGaugh, and Schombert 2016) provides observed rotation curves and baryonic mass profiles derived from 3.6 m Spitzer photometry for 175 nearby disk galaxies. After applying standard quality filters (inclination , distance Mpc, sufficient radial coverage), 137 galaxies remain in the test sample. For each galaxy, we compute the predicted rotation curve using only the observed baryonic mass profile and a single universal acceleration scale — no per-galaxy fitting, no halo parameters, no adjustable constants.

The predicted gravitational acceleration at radius is:

where is the Newtonian acceleration computed from the baryonic mass profile and is an interpolation function that transitions between the Newtonian regime (for) and the weak-field SFT regime (for). The interpolation function is motivated by the screen-energy construction rather than introduced as a purely empirical MOND ansatz; the full derivation and boundary-condition analysis are deferred to the companion. Three-sentence form:

Starting point. The local information-maintenance cost inside a galaxy is governed by a screen-energy functional that measures how much of the Compton-maintenance budget is absorbed by the pre-existing gravitational curvature versus transmitted to the cosmological boundary.

Pivot. The effective gravitational acceleration at a given radius is the gradient of the screen-energy functional, and requiring the screen to be self-consistent with the local baryonic mass distribution yields a closed equation for in terms of the acceleration ratio .

Result. The interpolation function takes the form

where is a morphology-dependent coefficient determined by the projection geometry of the screen onto the local matter distribution. The companion document presents the full screen-energy derivation and the boundary-condition analysis that fixes for each morphological class.

The comparison table. To evaluate the physical significance of the factor, we compare four acceleration scales against the same SPARC sample using the same interpolation function and the same quality filters. Only the acceleration scale differs between rows.

Model	Mean relRMSE	Improvement
Baryons only (no correction)	0.458	—

Model	Mean relRMSE	Improvement
SFT: m/s	0.167	63% reduction
Alternative: (no factor)	0.283	38% reduction
Alternative: m/s	0.174	62% reduction

Table 1: Rotation curve error across 137 SPARC galaxies for four candidate acceleration scales. The theoretically derived scale outperforms all alternatives, including the literal scale . The interpolation function and quality filters are identical across all four rows; only the acceleration scale differs.

The table confirms two independent claims simultaneously. First, that the Hubble-scale constant sets the relevant weak-field acceleration scale — eliminating baryons-only models by a factor of 2.7 in error. Second, that the specific factor of is physically real — the derived scale outperforms the literal by a factor of 1.7 in error. The data independently select the factor that the Compton Maintenance Postulate predicts, and they do so before any theoretical derivation has been invoked. This is the load-bearing rhetorical point of the SPARC test: the fact that the comparison exists and that the data prefer the derived value is what distinguishes this result from a fit.

The close performance of (relRMSE 0.174 versus 0.167 for) reflects the numerical proximity of to . The 137-galaxy sample prefers the derived value but does not exclude nearby alternatives at high statistical significance. Future high-precision surveys (WALLABY, SKA, and the Roman Space Telescope) will provide larger samples and lower systematic uncertainties and are expected to sharpen this discrimination considerably.

Morphology dependence. A more precise test comes from examining how varies across galaxy morphology. The screen-energy construction predicts that is not universal but depends on the projection geometry of the local screen onto the baryonic mass distribution. In the current framework, the two load-bearing reference values are an isotropic spherical-screen limit and a flat-disk axial-projection limit. The spherical limit is the more conservative derived reference point; the disk-sector value is the leading current candidate emerging from the projection-integral program. The main paper treats the morphology dependence as empirically established and the exact morphology-to- mapping as an open derivation problem whose full treatment belongs in the companion.

Numerically, the two benchmark values carried by the current weak-field program are:

The SPARC data support a morphology dependence in this range, with diskier systems favoring the lower benchmark and bulge-richer systems favoring the higher one. What remains open is not whether morphology matters — the data already say that it does — but the first-principles derivation of how a galaxy's observed mass distribution selects its effective screen geometry.

This morphology dependence is confirmed against the SPARC sample. Using bulge fraction as the morphological covariate and controlling for additional morphological and photometric variables through a partial-correlation test, the coefficient correlates with bulge fraction at .

The full partial-correlation analysis and the galaxy-by-galaxy classification are presented in the companion document.

The structural link to cosmology. The projection integrals that produce the two benchmark values at galactic scales are the same integrals — applied at a different scale — that produce the static projection corrections and for the four-state cosmic web ladder in Section 10. This is the first appearance of a structural unification that the paper will develop in two stages: one projection-integral mechanism, applied at galactic scales, produces the morphology-dependent values that fit the SPARC data; the same mechanism, applied at cosmic-web scales, produces the wall and filament projection corrections that structure the late-universe dark energy equation of state. This is not a coincidence and it is not an interpretation. It is an algebraic consequence of the same screen-geometry construction operating at two different scales, and Section 10 will show the calculation explicitly.

What this section establishes. The galactic acceleration threshold is a falsifiable prediction of the Compton Maintenance Postulate. It has been tested against 137 SPARC galaxies with zero free parameters and survived at 63% error reduction. The morphology dependence of has been tested against the same sample and confirmed at . The data prefer the derived factor over the literal alternative, confirming Claim 2 of the Compton Maintenance Postulate (one bit equals one complete cycle, not one radian). The framework is now empirically anchored at galactic scales, and the structural link between the galactic projection integrals and the cosmological projection corrections provides the scaffolding for the cosmological results of Section 10.

7. The Structural Identity

Section 5 produced a modified Friedmann equation differing from the standard result by a content-independent factor of exactly . Section 6 showed that the square root of this factor at galactic scales produces an acceleration threshold that the SPARC data confirm at 63% error reduction. This section makes the structural argument that ties these results together with three further appearances of the same geometric identity. The factor is not a single unexplained discrepancy. It appears in five distinct physical contexts, and in each case it decomposes into the same two independent factors of — one from the phase convention (linear cycles versus angular radians) and one from the boundary-versus-bulk normalization (Bekenstein-Hawking horizon counting versus Einstein field equation coupling). Four of these appearances are derived directly from the Compton Maintenance Postulate. The fifth, in the cosmic microwave background transfer coefficient, is a candidate splice at candidate-derivation status — numerically viable and physically motivated, but not yet closed as a theorem.

The structural claim is that this recurrence is the signature of a single underlying mechanism rather than a sequence of numerical coincidences. Each appearance has a distinct physical context, but each decomposes into the same two factors for the same reason. The section develops each appearance in turn, then summarizes the unification in Table 2.

The two-factor decomposition. Before listing the appearances, we state the decomposition explicitly:

The first factor of : phase convention. The Compton Maintenance Postulate identifies one irreversible bit with one complete phase cycle (radians) rather than one radian of phase advance. All SFT formulas involving bulk information processing therefore use (linear Planck constant, counting complete cycles) rather than (reduced Planck constant, counting radians). Since , every equation that contains a ratio between bulk processing and boundary accounting inherits a factor of from this choice. This factor is empirically confirmed by the SPARC comparison in Section 6: the data prefer over , which is the direct numerical signature of the linear-versus-angular phase convention.

The second factor of : boundary-bulk normalization. The Bekenstein-Hawking entropy formula counts boundary cells as , where the factor of 4 is the thermodynamic normalization that sets the number of independent degrees of freedom on a gravitational horizon. The Einstein field equation couples matter to geometry through the coefficient , where the is not derived within general relativity itself but is fixed by requiring consistency with the Newtonian limit via the Poisson equation . The ratio between these two normalization conventions — how SFT counts boundary cells versus how general relativity couples bulk geometry to matter — introduces the second factor of . This factor is identified algebraically in the current paper but is not yet derived from first principles; closing this gap is Open Theorem 2 in Section 14.

Appearance 1: The Friedmann ratio. As derived in Section 5, the ratio of the SFT modified Friedmann equation to the standard Friedmann equation is:

Both factors enter. The first factor of enters through the Compton maintenance rate , which uses rather than . The second factor of enters through the comparison between the Bekenstein-Hawking boundary counting on the SFT side and the Einstein coupling on the standard Friedmann side. The density cancels, the gravitational constant cancels, and what remains is the pure geometric factor , content-independent and universal. This is the foundational appearance from which the other four follow.

Appearance 2: The galactic acceleration scale. The acceleration threshold derived in Section 6 is:

Only one factor of appears at the galactic scale, not the full . The reason is that acceleration scales linearly with while the cosmological ratio scales with . Taking the square root of exposes only one factor of the ratio, and that factor is what governs the galactic transition. This is why Milgrom's empirical is close to rather than or : the universe's weak-field acceleration threshold is the square root of the cosmological factor, measured at the present epoch. The SPARC test (Section 6) confirmed this specific factor against three alternative scales and found that the data prefer . Appearance 2 is therefore empirically confirmed, not merely algebraically derived.

Appearance 3: The holographic capacity gap. Section 2 described the apparent over-provisioning of the cosmological boundary: the Bekenstein-Hawking entropy of the Hubble horizon (bits) exceeds the total astrophysical processing history of the universe (operations) by approximately nineteen orders of magnitude. SFT shows that this gap is a bookkeeping

artifact produced by the mismatch between angular and linear counting conventions. Three-sentence form:

Starting point. The total number of Compton cycles performed by all mass-energy in the observable universe over its entire history is

which counts one tick per complete Compton cycle per particle, integrated over the Hubble time. The total number of Planck-area cells on the Hubble horizon is

which counts one bit per Planck area on the horizon, as prescribed by the Bekenstein-Hawking formula.

Pivot. These two counts are being measured in inconsistent units. uses (linear cycles), while uses (which depends on , angular phase). The apparent discrepancy is therefore an artifact of the phase convention, not a physical over-provisioning.

Result. The ratio of the two counts is

The factor of that separates them is exactly the same structural identity that appears in the Friedmann ratio, decomposing into the same two factors of for the same two reasons. The holographic boundary is not over-provisioned; it is being counted in angular units while the bulk processing is being counted in linear units. If both counts are restated in consistent units — either all angular or all linear — the apparent excess disappears entirely. The nineteen-order-of-magnitude discrepancy that Section 2 flagged as an unsolved problem is resolved structurally, without invoking any new physics, as the same identity viewed from a different angle.

Appearance 4: The dark energy density structural gap. The matter-only SFT modified Friedmann equation gives, for :

leaving a structural gap of approximately 44% between the matter-only SFT prediction and the observed expansion rate. This gap is not covered by the bare Friedmann ratio of Appearance 1; it is the remaining space that the cosmological dynamics of Sections 9 through 11 must fill. The claim of SFT is that this gap is not the signature of a cosmological constant or a dark energy fluid, but the structural signature of the same identity manifesting through the forced thermodynamic transitions of the maintenance process. The detailed closure is developed in Section 10 via the four-state cosmic web ladder; what Appearance 4 establishes is that the *size* of the gap is structured by , not by a free parameter. The density and the gravitational constant have already canceled in the ratio; what determines how the gap is filled is the dynamics of forced transitions, not an additional fitting degree of freedom.

Appearance 5: The cosmic microwave background transfer coefficient. *(Candidate-derivation status.)* The filtered 3D-to-2D projection of a relic phase perturbation onto the last-

scattering surface produces an observable signature governed by a transfer coefficient . Numerical sweeps across plausible parameter ranges identify the required value of in the neighborhood of , and partition tests rule out simple abundance ratios as explanations. The compound transfer law that fits the data takes the form:

where is the temperature-conversion factor from the Stefan-Boltzmann relation , and are the matter and radiation content fractions at the relevant epoch, and is a residual efficiency factor. The empirically preferred value of is approximately . The SFT candidate splice identifies this residual with the structural factor:

which produces the candidate transfer law:

Parameter points that reproduce the target value include giving , giving , and giving . The numerical match is tight across plausible parameter ranges, and the splice produces 17 ballpark hits in a broader sweep where simple partition alternatives (alone, alone) produced zero.

The physical interpretation is that the factor encodes an angular-to-linear phase-space conversion. The internal Compton clock ticks in an angular phase space bounded by per cycle. The CMB observable is a linear thermodynamic measurement in a three-dimensional spatial volume that projects onto a two-dimensional last-scattering surface. Projecting an internal angular variance onto an external linear space geometrically requires dividing by the product of the two angular domains, which is . The full derivation of this conversion — writing the unbroken symbolic chain from the relic phase perturbation through the photon-fluid response to the observed temperature anisotropy, with the factor attached to an explicit angular-to-linear conversion step — is the work of Section 11, which presents the candidate splice at its current epistemic status. The closure of this derivation is Open Theorem 3 in Section 14.

The epistemic status of Appearance 5 must be stated explicitly. Appearances 1 through 4 are derived algebraically from the Compton Maintenance Postulate and the master equation; they follow within the stated axiomatic scaffold. Appearance 5 is a candidate splice: the factor is numerically preferred over simpler alternatives, it has a physically motivated interpretation as an angular-to-linear phase-space conversion, and it appears in a context developed independently of the cosmological and galactic sectors. But the unbroken symbolic derivation is not yet in hand. Until it is, Appearance 5 is included in the unification not as a fifth derived appearance but as a fifth candidate appearance, and the paper is explicit about the distinction. This is the honest epistemic register, and it is the register the main paper will maintain for this result throughout.

A further open question: whether the angular-to-linear interpretation of Appearance 5 is the same factor as the phase-convention decomposition that enters Appearances 1 through 4, or a consistent extension of it, or a genuinely different that happens to share the numerical value. The current paper takes no position on this question and flags it explicitly as part of Open Theorem 3.

The unification table.

#	Physical context	Factor	SFT side	Standard side	Status
1	Friedmann ratio				Derived
2	Galactic acceleration scale			(no standard prediction)	Derived & confirmed
3	Holographic capacity gap		(-based)	(-based)	Derived
4	Dark energy structural gap		matter-only gap		Structurally implied
5	CMB transfer coefficient			(no standard prediction)	Candidate splice

Table 2: Five appearances of the structural identity across distinct physical contexts. Appearances 1 and 3 are algebraically derived; Appearance 2 is derived and empirically confirmed against the SPARC rotation curve sample; Appearance 4 is structurally implied by the Friedmann ratio and closed dynamically in Section 10 via the four-state ladder; Appearance 5 is a candidate splice at candidate-derivation status, with a physically motivated interpretation as an angular-to-linear phase-space conversion and an unbroken symbolic derivation pending (Open Theorem 3).

What this section establishes. The factor is not a single unexplained constant. It appears in five distinct physical contexts — four derived and one candidate — and in each context it decomposes into the same two independent factors of for the same two reasons. Four of the five appearances are forced by the Compton Maintenance Postulate and the axiomatic foundation of Section 3; the fifth is a numerically viable candidate whose derivation is the subject of ongoing work. The structural recurrence is the signature of a single underlying mechanism: the geometric cost of mapping angular quantum phase evolution onto a linear holographic boundary, measured at the scale where the quantum and gravitational accounting systems interface. The rest of the paper uses this structural identity as a spine: Section 8 shows that the master equation promotes cleanly to a variational principle, Sections 9 through 11 develop the dynamical consequences via forced thermodynamic transitions, and Section 12 closes out the gravitational sector by recovering exact general relativity in the strong-field limit.

8. The Holographic Count Equation and the Constrained Action

Section 5 derived the modified Friedmann equation algebraically from the master equation together with the Compton Maintenance Postulate. That derivation is sufficient for the

cosmological prediction, but it does not by itself display the framework in variational form or explain why one Hubble time is the natural settlement interval for the information ledger. This section therefore introduces the holographic count equation and shows that the same FLRW background law can be enforced through a constrained minisuperspace action. In that sense, the count equation supplies the native variational scaffolding of the framework, even though the full Euler-Lagrange development is deferred to the companion derivations.

Motivation. The algebraic derivation of Section 5 shows that follows from , , and . This is sufficient to produce a testable prediction, and the SPARC confirmation of Section 6 establishes that the prediction survives empirical test at the galactic scale. But algebraic derivation is not the same as variational formulation. A proper dynamical theory should specify what is being extremized and why — what the action is, what the equations of motion are, what constraints are enforced and by what mathematical mechanism. Without this structure, the theory is a relation between observables rather than a set of field equations.

An earlier formulation of the theory (discussed in the companion document) attempted to write SFT as an instantaneous rate law of the form , where was an unexplained constant that had to be hand-set to the value 3 in order to recover the correct coefficient in the modified Friedmann equation. The requirement was a symptom of a missing structural ingredient: the instantaneous rate law was attempting to impose the count balance moment by moment without accounting for the causal timescale over which the balance can be physically established. The holographic count equation resolves this by recognizing that the information ledger is not settled instantaneously but over one Hubble time, and the factor of 3 emerges automatically from the ratio between the Hubble sphere's volume and its surface area rather than being inserted as a free parameter.

The holographic count equation. The statement is that the boundary's information capacity at a given moment equals the total bulk processing accumulated over one Hubble maintenance interval:

The left side counts the number of Planck-area cells on the Hubble horizon. The right side counts the number of complete Compton phase cycles performed by all bulk mass-energy during one Hubble time . The equation asserts that these two numbers are equal at every moment of cosmic time: the information capacity of the boundary exactly matches the information processed by the bulk over one causal maintenance interval.

The choice of the Hubble time as the settlement interval is not arbitrary. It is the timescale over which a signal can traverse the Hubble radius , and therefore the unique interval within which bulk information processing can be causally correlated with the boundary capacity that accommodates it. Beyond one Hubble time, the ledger cannot be settled — regions of the bulk fall out of causal contact with the accommodating boundary, and the balance becomes ill-defined. The count equation is therefore an apparent-horizon thermodynamic balance, in the sense developed by Padmanabhan (2010) and Jacobson (1995): the boundary refreshes its information budget every Hubble time, and the bulk fills it exactly over that interval.

Derivation of the modified Friedmann equation from the count equation. Three-sentence form:

Starting point. The left side of the count equation is , using . The right side is , where is the total energy within the Hubble volume and is the product of the Planck constant and the Hubble rate.

Pivot. Setting the two sides equal and substituting the explicit expressions:

Substituting on the left side and simplifying, the factors of cancel, the factor of cancels, and one factor of cancels.

Result. What remains is

which is exactly the modified Friedmann equation derived algebraically in Section 5, now obtained from the holographic count equation with zero proportionality constants and no hand-set parameters. The factor of 3 that appeared as an unexplained in the earlier instantaneous rate law emerges automatically from the algebra: it is the ratio between the Hubble sphere's volume (where ticking occurs) and its surface area (where capacity is measured), encoded in the factor of in the volume and the factor of in the area. The count equation was never missing a constant; the earlier formulation was missing the causal interval that makes the ratio well-defined.

Promotion to a constrained action. The count equation can be enforced as a constraint through a Lagrange multiplier , producing a constrained minisuperspace action:

where is the FLRW scale factor, , and is the standard matter action that ensures the continuity equation for pressureless dust. The FLRW variation of this action reproduces the same background modified Friedmann equation, with the full variational details deferred to the companion derivations.

This is not a standard local Lagrangian density integrated over spacetime. It is a constrained minisuperspace action — a holographic constraint enforced at the boundary level, relating the rate of boundary growth to the rate of bulk information processing. The fundamental law of the theory is a balance between surface capacity and volume content, consistent with the holographic and information-theoretic character of the framework. The Lagrange multiplier plays the role of a maintenance-cost enforcer: it measures the cost of violating the count balance, and at the classical level its value is determined by the constraint itself and does not introduce independent dynamics. This is directly analogous to the role of the lapse function in canonical general relativity, where the Hamiltonian constraint is enforced by a multiplier that does not generate new physical degrees of freedom.

Summary table of formulations.

Formulation	Equation	Free parameters	Recovers ?
Master equation (algebraic)		None	Yes

Formulation	Equation	Free parameters	Recovers ?
Instantaneous rate law		(hand-set)	Only if
Holographic count equation	(over)	None	Yes (derived)
Constrained action	with	None	Yes (variational, companion)

Table 3: Four formulations of the SFT cosmological law, in order of increasing mathematical maturity. The algebraic derivation of the master equation is sufficient to produce testable predictions but does not specify a variational principle. The instantaneous rate law, attempted as an intermediate step, required to be hand-set and was structurally incomplete. The holographic count equation and the constrained action are the native variational formulations of the theory; both produce the modified Friedmann equation with zero free parameters.

What this section establishes. Solvency Field Theory has a native constrained minisuperspace formulation. The master equation , the modified Friedmann equation , the holographic count equation , and the constrained minisuperspace action with Lagrange multiplier are four expressions of the same physical content at increasing levels of mathematical maturity. Each produces the same modified Friedmann equation with zero free parameters. The constrained action formulation establishes that the modified Friedmann law can be embedded in a standard variational scaffold rather than appearing only as an algebraic coincidence, and it is the formulation that would be extended in the open research direction of generalizing SFT beyond FLRW minisuperspace to a full field-theoretic formulation (Open Theorem 6 in Section 14).

9. Forced Thermodynamic Transitions of the Maintenance Process

Sections 3 through 8 established the foundational layer of Solvency Field Theory — the axioms, the master equation, the Compton Maintenance Postulate, the modified Friedmann equation, the galactic confirmation, the five-appearance structural identity, and the variational principle. Everything so far is static, in the sense that it describes what the maintenance process is and what it requires at any given moment of cosmic time. This section introduces the dynamical claim that structures the rest of the paper: the maintenance process has thermodynamic viability conditions, and when a viability condition is violated, the system undergoes a forced transition to a new maintenance regime. Forced transitions leave observable scars because they alter the geometric response of the ledger to bulk processing. Two such transitions — one at the early universe, one at the late universe — produce the cosmic microwave background imprint and the dark energy phantom crossing respectively. This is the general principle. Sections 10 and 11 deliver the two instances in quantitative detail.

The general principle. The Compton Maintenance Postulate establishes that every massive particle processes information continuously at its Compton frequency and that each complete cycle demands one Planck area of boundary accommodation. The master equation requires that the cosmological geometry expand at exactly the rate needed to accommodate bulk processing. Together, these determine a specific maintenance regime at each moment of cosmic

history: a specific rate of information production , a specific boundary capacity , and a specific expansion rate that satisfies the balance.

But the maintenance regime is not unconditional. It depends on the physical state of the universe — on temperature, on the available channels of entropy production, on the fraction of mass that is gravitationally organized, on the balance between active transactional processing and passive structural occupancy. As the universe evolves, these state variables change. At some moments the change is smooth: the current maintenance regime remains thermodynamically viable, and the system adjusts continuously. At other moments the change is not smooth: the current regime becomes thermodynamically invalid, and the system is forced to transition to a new regime that can accommodate the altered state.

A forced transition is a specific thermodynamic event. It is the moment at which a maintenance budget — a ratio measuring the ledger's ability to sustain its current processing — crosses a critical threshold, rendering the current regime unable to balance bulk production against boundary accommodation. When this happens, the system has no continuous path forward: it must undergo a discontinuous or rapid reorganization of its maintenance structure. The reorganization produces an observable signature because the geometric response of the ledger to bulk processing changes abruptly. What we observe as a scar on the cosmic record is the imprint of a maintenance regime that could no longer hold.

The mathematical framework for describing forced transitions is familiar from condensed-matter and statistical physics. A maintenance budget can be written generically as:

where is a control parameter (temperature, structural fraction, or any other state variable that drives the transition), is the rate at which the current regime can accommodate bulk processing through shared channels, and is the maximum rate the regime can physically sustain. The critical condition defines the transition: below the threshold the regime is viable, above it the regime cannot balance production against accommodation. Near the critical point, the transition can often be described in Landau form with a control parameter and an order parameter that emerges as for . This is the mildest possible scaffolding for a thermodynamic phase transition, and it is the scaffolding that SFT inherits naturally from its axiomatic foundation: Landauer cost (Axiom 1) establishes the thermodynamic dimension, thermal time (Axiom 2) establishes that the transition has a duration, holographic capacity (Axiom 3) establishes the ceiling that the budget is measured against, and the maintenance postulate (Axiom 4) establishes that expansion responds to the budget at every moment.

Two instances, one principle. The rest of this section describes the two forced transitions that Sections 10 and 11 develop quantitatively. The order is reverse-chronological — the late-universe transition first, then the early-universe transition — because the late-universe side is more empirically developed and provides the clearer template for understanding what a forced transition looks like.

The late-universe engine crash. As the universe ages, the active astrophysical entropy engines — star formation and supermassive black hole accretion — deplete the baryonic gas reservoirs that fuel them. The IllustrisTNG TNG300 simulation, used as a calibrated proxy for the thermodynamic history of the observable universe, shows that both engines crash in near-

lockstep by approximately 70% between and , with the steepest decline centered near . The active-dominated maintenance regime, in which the universe's information budget is serviced primarily by dissipative baryonic processing, becomes unable to supply the ledger at the rate required to service the accumulated information budget. The maintenance budget crosses its critical threshold, and the system is forced to transition to a structure-dominated regime in which the passive occupancy cost of organized gravitational architecture — virialized halo mass — dominates the maintenance balance. The transition produces an observable signature: the dark energy equation of state evolves from (active-dominated, matter-like) toward (structure-dominated, phantom-like), with the specific evolution measured by DESI DR2. The engine crash is the *informational cause* of what the DESI survey is optically detecting as the dark energy phantom crossing. Section 10 develops the full quantitative framework, including the four-state cosmic web ladder that replaces the earlier bimodal decomposition as the current cosmological formulation.

The early-universe shatter. As the universe cools below matter-radiation equality, the ledger is forced to track an enormous new population of independent Compton clocks. In the radiation-dominated epoch, most of the universe's mass-energy was in relativistic species whose maintenance could be bookkept through a coupled thermal processing regime — one regime managing the entire radiation fluid as a shared bookkeeping system. As temperature drops through the matter-radiation handoff, massive particles (predominantly baryons, but also cold dark matter if present) transition from contributing to the shared radiation regime to demanding independent Compton maintenance, each particle requiring its own continuous phase evolution at its own Compton frequency. The coupled maintenance regime becomes thermodynamically invalid: the maintenance budget crosses unity at a critical temperature , and the system is forced to transition to an independent-clock regime. Landau-form analysis of the transition suggests that the order parameter scales as near the critical point, and mild parameter scans place in the 5000–10000 K neighborhood — the thermodynamic neighborhood of recombination but not necessarily identical to the optical decoupling temperature of 3000 K. This is the forced transition that produces the relic phase perturbation field whose 3D-to-2D projection onto the last-scattering surface is observed as the cosmic microwave background anisotropy. Section 11 develops the quantitative framework.

Structural unification: one mechanism, two epochs. The claim of SFT is that the late-universe engine crash and the early-universe shatter are two instances of the same underlying physical principle — forced thermodynamic transition of the maintenance process — operating at opposite ends of cosmic history and at opposite ends of the control parameter. The engine crash is driven by structural exhaustion: as the universe ages, the active processing channels run out of fuel, and the system is forced to reorganize around passive structural occupancy. The shatter is driven by thermal cooling: as the universe ages, the temperature drops through a critical value, and the system is forced to reorganize from shared to independent maintenance regimes. The control parameters are different (structural fraction versus temperature), the critical values are different, and the detailed dynamics are different. But the underlying logic is identical: a maintenance regime becomes thermodynamically invalid, and the system undergoes a forced reorganization that leaves a geometric scar on the cosmic record.

The structural unification is claimed at the level of physical principle and at the level of mathematical scaffolding. Both instances are governed by a maintenance budget that crosses a

critical threshold. Both instances can be described in Landau form near the critical point. Both instances produce observable signatures through the altered geometric response of the ledger. The two transitions are mechanistically parallel, and recognizing them as instances of one principle is what makes SFT's early-universe and late-universe sectors structurally coherent rather than two independent applications of the master equation.

What structural unification does and does not claim. The formal unification — a single closed equation whose two solutions are the engine crash and the shatter — is not yet in hand. Writing that equation is Open Theorem 1 in Section 14. The candidate approach is to generalize the maintenance budget to a function of both temperature and structural fraction, such that the critical condition has two distinct solutions at opposite ends of cosmic history, and to show that the generalized budget reduces to the shatter equations at early times and the engine-crash equations at late times. This generalization is the next theorem the framework is working toward. Until it is closed, the unification is structural rather than formal: the two instances share a physical principle and a mathematical scaffolding, but the detailed equations currently live in two separate subsystems. The main paper is explicit about this distinction, and it does not overclaim.

What this section establishes. The maintenance process has thermodynamic viability conditions, and when a viability condition is violated, the system undergoes a forced transition to a new maintenance regime. Two such transitions — the late-universe engine crash and the early-universe shatter — produce the observable signatures that Sections 10 and 11 develop quantitatively. The two transitions are instances of one physical principle, structurally unified at the level of mechanism and mathematical scaffolding, with formal unification flagged as an open theorem. This is the dynamical claim that carries the paper's cosmological and early-universe sectors, and it is the claim that transforms SFT from a static framework (the master equation and its consequences at each moment) into a dynamical one (the evolution of the maintenance process across cosmic history and the scars that evolution leaves).

10. The Cosmological Sector: The Four-State Ladder and the Dark Energy Equation of State

Section 9 established that the late-universe engine crash is a forced thermodynamic transition of the maintenance process from an active-dominated regime to a structure-dominated regime. This section delivers the quantitative framework. The cosmic web decomposes into four distinct environments — voids, walls, filaments, and virialized halos — and each environment has a different maintenance regime because each has a different local screening capacity. The decomposition is not phenomenological: within the current framework, it is motivated as the minimal ladder consistent with three structural axioms applied to the geometry of the cosmic web. The section presents the four-state ladder as the leading current structural closure of the cosmological sector, derives the static projection corrections from the same spherical-screen integrals that produced the galactic values in Section 6, presents the dimensionally corrected chain from first principles, and situates the historical cosmology pipeline that produced the quoted DESI-facing CPL result alongside the corrected chain with an explicit audit status for V10.1 reconciliation.

The cosmic web and its four states. Large-scale structure observations and high-resolution cosmological simulations agree that the mass distribution of the universe at late times is organized into a cosmic web of four distinct environmental categories. Voids are under-dense regions containing approximately 15% of the cosmic mass in 80% of the cosmic volume. Walls are two-dimensional sheets bounding voids, containing approximately 25% of the mass in 12% of the volume. Filaments are one-dimensional structures connecting higher-density nodes, containing approximately 35% of the mass in 6% of the volume. Virialized halos are three-dimensional collapsed gravitationally bound structures, containing approximately 25% of the mass in 2% of the volume. These numbers are approximate and depend on the identification algorithm, but the four-category structure is robust across all modern cosmological analyses.

Each of the four environments has a different local gravitational infrastructure. Halos are fully virialized, with a local gravitational curvature that can absorb the Compton maintenance cost entirely at the local scale — the particle's information budget is settled against pre-existing geometry, and the maintenance cost is transmitted to the cosmological boundary only minimally. Filaments have partial gravitational organization along one axis (the filament's long direction) but not the other two, so they can absorb the maintenance cost along one dimension but transmit it along the others. Walls have partial organization in two dimensions but not the third. Voids have essentially no local gravitational organization, and each particle's maintenance cost is transmitted in full to the cosmological boundary.

The four-state ladder as a structural result. SFT's cosmological sector treats this environmental decomposition not as a phenomenological labeling but as a structural result following from three stated axioms. The decomposition is not introduced as an arbitrary fit; within the current framework it is motivated as the minimal ladder consistent with three structural axioms about neutrality, axis factorization, and consistency with the identity:

Axiom C1 (volume-weighted neutrality): The total Compton maintenance demand of the universe, summed over all environments weighted by their volume fractions, must equal the total maintenance demand computed at the mean density. This is a conservation statement — the cosmic web redistributes maintenance across environments, but the total is fixed.

Axiom C2 (axis factorization): The local maintenance cost transmitted to the cosmological boundary factors over the independent spatial axes of the local environment. A halo is three-dimensionally organized, so no axes transmit. A filament is organized along one axis (longitudinal), so two axes transmit. A wall is organized along two axes, so one axis transmits. A void is organized along zero axes, so all three axes transmit.

Axiom C3 (consistency with the structural identity): The maintenance cost transmitted from an environment to the boundary, normalized to the fully unscreened case, must be expressible through powers of the same structural factor that appears throughout the framework.

From these three axioms, the environmental maintenance weights follow. Three-sentence form:

Starting point. Let denote the maintenance cost transmitted from environment "env" to the cosmological boundary, normalized such that (full transmission through all three spatial axes in an unscreened void) and (no transmission, complete local screening). The axis-factorization

axiom requires that the cost transmitted through independent axes is (cost per axis), where the cost per axis is a fixed geometric factor.

Pivot. Consistency with Axiom C3 requires that the per-axis cost be , such that three-axis transmission (void) gives , and zero-axis transmission (halo) gives . The intermediate cases are determined: one-axis transmission (filament) gives , and two-axis transmission (wall) gives .

Result. The four-state ladder is:

These are not introduced as fitted weights. Within the stated axioms they are the natural ladder values; whether that ladder is strictly unique in a stronger mathematical sense is a question better handled in the companion derivation document. The void state sits at the full factor because voids transmit through all three axes; the halo state sits at unity because halos are fully screened; the filament and wall states sit at intermediate values determined by the factorization.

Static projection corrections from spherical-screen integrals. The ladder values above are the *ideal* weights — the values for environments whose internal geometry is rigid and whose orientation is averaged over all directions. Real filaments and walls have finite thickness and anisotropy that modify the transmitted cost through additional projection factors. The computation of these projection factors uses the same spherical-screen integrals that produced the morphology-dependent values in Section 6. Three-sentence form for the wall correction:

Starting point. A wall is modeled as a flat sheet of finite thickness embedded in a three-dimensional background, with the screen energy functional integrated over the sheet's geometry. The relevant integral is the second-moment of the spherical-screen projection onto the wall's normal axis.

Pivot. Evaluating the second-moment integral for a flat-sheet geometry yields a projection factor of relative to the ideal wall value, because only one-third of the screen's angular support projects onto the normal direction when averaged over all incidence angles.

Result. The static projection correction for the wall state is

and the corrected wall weight is .

The same analysis for a one-dimensional filament yields:

because two-thirds of the screen's angular support projects onto the two transverse directions. The corrected filament weight is .

Structural unification with the galactic sector. The spherical-screen second-moment integrals that produced and at cosmological scales are the same integrals that produced the benchmark values at galactic scales in Section 6. This is the structural unification that Section 6 anticipated: one projection-integral mechanism, applied at two different scales, produces four

distinct observable signatures — two morphology-dependent values for galactic rotation curves and two environment-dependent values for cosmic-web maintenance weights. The underlying mathematical object is the same in both cases: the angular projection of a spherical screen onto a geometry of lower dimensionality. At galactic scales, the screen projects onto disk or bulge geometries and produces . At cosmological scales, the screen projects onto wall or filament geometries and produces . The fact that both calculations use the same integral and produce geometrically sensible results at both scales is a structural confirmation that the screen-energy construction is operating consistently across the weak-field regime of the theory. The full derivation of both the galactic and cosmological projection integrals, including the boundary-condition analyses that distinguish the spherical, axial, wall, and filament cases, is in the companion document.

The corrected chain from first principles. To connect the four-state ladder to the observable dark-energy equation of state, we need the mapping from the master equation to an effective equation of state that can be compared to DESI measurements. The derivation below separates three distinct objects that were conflated in earlier formulations of the framework: the instantaneous expansion law, its time evolution, and the effective equation of state reconstructed from that evolution. The key structural point is that an equation of state depends on how the expansion rate *changes*, not merely on the instantaneous ratio . Any correct expression must therefore involve , or equivalently the time derivatives of and . The derivation proceeds in three steps.

Step 1: Differentiate the master equation. Starting from , differentiate with respect to cosmic time:

This step is exact. It shows that is sourced by the difference between the fractional rate of change of the information production rate and the fractional rate of change of the boundary capacity. Both and have units of inverse time, and their difference is what drives the expansion history to deviate from constant.

Step 2: Apply the standard flat-FLRW reconstruction identity. The effective total equation of state corresponding to a given expansion history in a flat FLRW spacetime is:

This is a standard identity in general relativistic cosmology: it is how any observationally reconstructed expansion history is converted to an effective fluid equation of state, and it holds independently of the underlying physical model driving . Substituting the master-equation form of from Step 1:

This is the dimensionally consistent expression for the total effective equation of state within SFT. It depends on the time evolution of the maintenance balance — and — not on the instantaneous ratio alone. A constant maintenance history (,) yields , which mimics a cosmological constant. A declining active-maintenance channel drives , which pushes below into phantom-like behavior. The forced transitions of Section 9 are precisely the events at which changes discontinuously, and the new expression makes visible in the math the

mechanism that Section 9 framed structurally: the equation of state is sensitive to how the maintenance regime is evolving, and forced transitions are exactly the moments when that evolution becomes observable.

Step 3: The dark-energy sector mapping. To compare SFT to DESI's dark-energy constraints specifically, one must isolate the effective dark-energy sector from the pressureless matter contribution. Let pressureless matter obey the standard continuity equation , and define the effective dark-energy density by

with the corresponding equation of state obtained by applying the continuity equation to this residual sector. The full symbolic mapping is developed in the companion; the main paper's load-bearing point is that the dark-energy-sector reconstruction follows the same logic as the total-fluid reconstruction above, with the matter contribution subtracted explicitly.

The historical cosmology pipeline. The cosmological result quoted in this framework — the dark energy equation of state , , with phantom crossing at , matching DESI DR2 at a distance of 0.005 in the CPL plane — was produced by a numerical pipeline that extracted the active and passive channel histories directly from the IllustrisTNG TNG300 cosmological simulation at ten snapshots from to , then mapped the resulting expansion history to a CPL dark-energy sector through an earlier working mapping. This historical pipeline is the empirical chain that produced the quoted DESI-facing numbers.

The active processing channel measures the rate of ongoing astrophysical transactions: black hole accretion rates computed as from the Bekenstein-Hawking scaling, and stellar formation rates computed from the standard Kennicutt-Schmidt relation. These represent the irreversible information-producing events in which matter is actively converting gravitational potential energy into thermal radiation or nuclear binding energy. The active channel peaks at and declines by approximately 70% to the present epoch as baryonic fuel reservoirs are exhausted.

The passive structural capacity channel measures the accumulated gravitational architecture: the virialized halo mass summed over all halos in each snapshot, weighted by the four-state ladder corrections. This represents the organized gravitational structure that has already been built and is now maintaining itself — the screening infrastructure that absorbs local maintenance costs. The passive channel grows monotonically from to as structure formation proceeds.

Applied to the TNG300 data through the historical pipeline with no cosmological fitting beyond the channel identification itself, the result is:

with phantom crossing at , matching the DESI-preferred crossing redshift. This is the result of an extensive numerical pipeline — many scripts, many intermediate calibrations, and a stable convergence across dataset variations — and it is the empirical output that anchors the framework's cosmological claims.

Audit status: reconciliation with the corrected chain. The dimensionally corrected derivation presented above is the mathematically preferred reconstruction of the effective

equation of state from the master equation. An earlier working formula used in the historical pipeline produced the quoted CPL pair through a different intermediate chain. A first coarse rerun of the corrected reconstruction, using a five-point channel history summary, preserves the crossing scale (, close to the historical 0.465) but shifts the fitted CPL coefficients substantially. The appropriate epistemic position is therefore:

The quoted CPL pair is the empirical output of the established historical cosmology pipeline. The corrected dimensionally consistent reconstruction from is the mathematically preferred framework-level derivation. Full reconciliation of the two pipelines — running the corrected reconstruction against the full original ten-snapshot channel history (or a denser one) with smoothed-derivative techniques appropriate to the corrected formula's sensitivity to — is flagged as an audit item for a V10.1 update. The qualitative result (phantom crossing at) is stable across both chains; the precise CPL coefficient reconciliation is the quantity under audit.

This is the honest status. The paper presents both chains: the historical pipeline with its well-established numerical result, and the corrected derivation with its mathematically clean framework-level form. The two are consistent at the crossing-redshift level and under audit at the CPL-coefficient level. A V10.1 update will either confirm full reconciliation or prompt a revision of Section 10's quantitative claims.

Load-bearing comparisons. Two comparisons from the historical pipeline are load-bearing for the main paper because they distinguish SFT's structure-dominated picture from generic dark-energy fits and from dark-matter phenomenology, and both results are stable independent of the specific chain used to map them to CPL coefficients.

The structural specificity test. When the passive channel is defined using raw dark matter halo mass instead of the virialized mass , the historical pipeline shifts the resulting equation of state to — overshooting the phantom divide and missing the observed value. The virialized mass , which specifically measures gravitationally organized, thermally relaxed structure, succeeds where raw inventory fails. This is physically meaningful within the SFT framework: the maintenance cost that matters is not the total amount of mass but the amount of mass that has been gravitationally organized into screening infrastructure. Unvirialized mass contributes to the gravitational inventory but does not participate in the same way in the information-maintenance balance. The distinction between and is a physical discriminator that no other dark energy model predicts, and it is the single strongest evidence that the passive channel is tracking real physics rather than fitting a phenomenological template.

The dynamical reservoir ablation. When the time-derivative term in the maintenance budget is removed (i.e., when the passive reservoir is treated as static rather than evolving), the DESI likelihood collapses by 36 to 1263 log-units depending on dataset. This is not a small effect — it is the difference between a strongly preferred fit and a catastrophically bad one. The ablation confirms that the dynamical maintenance machinery is doing real work: the evolution of the passive channel over cosmic time is not a decorative feature of the model, it is the mechanism by which the engine crash produces the observed phantom behavior. Both comparisons stay in the main paper because both are evidence that the structure-dominated picture tracks real physics.

CMB compatibility. The four-state ladder framework, when parameterized as a CPL dark energy model and propagated through the CLASS Boltzmann solver, produces a temperature

power spectrum that matches Planck 2018 at 0.078% RMS residual across multipoles –. The acoustic peaks are preserved to 0.104% in the first acoustic peak region (–) and 0.163% in the acoustic damping tail (–); the only region with noticeable deviation is the integrated Sachs-Wolfe sector at –, where the residual is 1.8% and consistent with cosmic variance at low multipoles. This compatibility is expected on physical grounds: at the CMB epoch (), the universe contains no virialized halos, no galaxy clusters, no voids, and no gravitational screening infrastructure. The Compton maintenance sector is subdominant because the universe is radiation-dominated and no organized structure exists to create differential screening. SFT's modifications emerge only at late times, as massive structures form and the screening mechanism activates. The framework preserves standard early-universe physics by construction.

Observer-side residual closure. A small residual in the low-redshift parameter remains after the four-state ladder is applied directly to the TNG300 data. This residual is closed through an observer-side local-environment correction based on the path-averaged unscreened burden along the line of sight from back to the observed redshift. The correction is derived from the covering geometry of the observer's local environment rather than fitted as a free parameter, and its robustness has been confirmed across a nine-point sensitivity sweep of the local environment parameters within a pre-committed DESI closure radius. The full derivation and the sensitivity sweep are in the companion document.

The bimodal model as coarse-grained limit. Earlier formulations of the cosmological sector used a bimodal (active/passive) decomposition rather than the four-state ladder. The bimodal model is not wrong; it is the coarse-grained limit of the four-state ladder in which voids and walls are consolidated into a single low-screening channel and filaments and halos into a single high-screening channel. The empirical success of the bimodal model in reproducing the DESI DR2 equation of state was the original signal that a fuller structural decomposition existed, and the four-state ladder is the resolution of that coarse-graining into its underlying environmental structure. This paragraph is the only mention of the bimodal model in the main paper; readers interested in the historical development of the cosmological sector can find the full account in the companion.

What this section establishes. The cosmological sector of Solvency Field Theory is anchored by the four-state ladder, presented as a structural result from three stated axioms and the identity. The static projection corrections and emerge from the same spherical-screen integrals that produced the galactic morphology parameters in Section 6, structurally unifying the galactic and cosmological sectors through one projection-integral mechanism. The dimensionally corrected chain from first principles — , , — provides the mathematically clean derivation from the master equation to the observable equation of state, with the dependence on making visible why forced transitions produce observable scars. The historical cosmology pipeline, which produced the quoted CPL pair with phantom crossing at matching DESI DR2, is carried alongside the corrected derivation, with full reconciliation between the two chains flagged as a V10.1 audit item. The structural specificity test and the dynamical reservoir ablation, both stable across chain choice, remain the load-bearing empirical evidence that the structure-dominated picture tracks real physics.

11. The Early-Universe Imprint: The Shatter and the CMB Transfer Coefficient

Section 9 established that the early-universe shatter is a forced thermodynamic transition of the maintenance process from a shared radiation-dominated regime to an independent-clock regime as the universe cools through the matter-radiation handoff. This section delivers the quantitative framework. The section is shorter than Section 10 because the early-universe side of the theory is less developed than the late-universe side, and the main paper is explicit about that asymmetry. The development proceeds in four stages: the minimal identity-like bridge connecting the cosmological sector to the relic phase field, the Landau-form analysis of the shatter transition, the 3D-to-2D projection of a relic phase event onto the last-scattering surface, and the candidate CMB transfer coefficient in which the fifth appearance of appears at candidate-derivation status. The epistemic register throughout this section is more tentative than Section 10, and this tentativeness is honest: the early-universe material is at candidate-derivation status, not closed theorem.

The minimal identity-like bridge. To connect the cosmological screening sector developed in Section 10 to the relic phase field that sources the CMB imprint, we need a bridge law linking the cosmological lag variable (where is the lag refractive index of the information-maintenance ledger), the screening fraction (where is the fraction of the cosmic volume screened by organized structure), and the relic phase field whose perturbations are observed in the CMB. The mildest possible identification — the version that minimizes commitments while still producing testable predictions — is the identity-like bridge:

This bridge identifies the perturbation in the lag variable with the perturbation in the relic phase field, takes the background phase rate to equal the FLRW count-equation clock , and relates the screening sector linearly to the lag variable through a single bridge coefficient . The bridge is not a postulated new field; it is the mildest identification consistent with the fact that both and measure the same underlying thing — the local rate at which the Compton maintenance process is advancing relative to the local boundary accommodation — at different epochs of cosmic history.

Mildness bound from recombination-era amplitudes. The identity-like bridge is quantitatively constrained by the amplitude of the relic phase perturbation at the recombination era. Using the standard relation between the phase perturbation and the observed temperature anisotropy:

with and , the recombination-scale amplitudes are at (horizon scale), at , and smaller beyond. Under the identity-like bridge, the perturbation in the screening sector is . A direct mildness scan across recombination-relevant wavenumbers yields:

These bounds establish that the identity-like bridge is not unconstrained. The bridge coefficient is quantitatively bounded on recombination-relevant scales, and the mildness bound falls within the range that generic structural expectations would place .

The Landau-form shatter. The forced transition from shared to independent Compton maintenance as the universe cools through the matter-radiation handoff can be analyzed in standard condensed-matter phase-transition language. Write a maintenance budget with a coherence overhead , where is a characteristic temperature and is a dimensionless exponent, and define the order parameter through:

This is the mildest Landau-form scaffolding that the transition admits, and it is the scaffolding that SFT inherits naturally from its axiomatic foundation. The critical temperature is exactly. The question is whether a plausible value can place the critical regime in the thermal neighborhood relevant to the CMB without forcing the relic phase amplitude to unphysical values.

Mild parameter scans with and shallow exponents or yield healthy solutions in the corner 5000–10000 K, giving relic phase amplitudes of order unity rather than pathological values. Steeper exponents are pushed out of the healthy region unless is strained. The mild solutions cluster in a surprisingly interpretable corner — shallow temperature dependence (linear or square-root in), characteristic temperature above but near the recombination-optical temperature, and coupling coefficients of order unity. The transition is structurally viable and mildly constrained, and it is the direct early-universe analog of the late-universe engine crash: a maintenance budget crossing a critical threshold, producing a Landau-form order parameter, and leaving an observable scar on the cosmic record.

3D-to-2D projection onto the last-scattering surface. The relic phase perturbation produced by the shatter is a three-dimensional field propagating through the early universe. The observable is a two-dimensional temperature anisotropy on the last-scattering sphere. The mapping from 3D source to 2D observable is a projection integral: the observed at a point on the sphere is some weighted integral of along the line of sight through the thin last-scattering shell. The form of this projection determines whether the shatter can produce a viable CMB imprint, or whether the 3D source amplitude must be so small that the shatter could not have occurred in any observable form.

The simplest source model is a spherical Gaussian relic event centered somewhere in the early universe:

with Laplacian . A thin-shell bridge proxy for the observed temperature anisotropy is:

where is the angular coordinate on the last-scattering sphere, is the 3D distance from the observer's sight line to the 3D source at each angular position, and is a transfer coefficient encoding the physical processes that convert the 3D phase perturbation into the 2D temperature observable.

Raw versus filtered projection. The raw projection of a Gaussian source onto the last-scattering shell, computed with , , , and (where is the distance from the observer to the source center and is the radius of the last-scattering sphere), produces a peak temperature anisotropy of . This overshoots the characteristic CMB amplitude of by a factor of approximately 3600. The 3D geometry of the source survives the projection — the angular width of the signal comes out near for the chosen , which is in the right ballpark for observed CMB features — but the raw amplitude is catastrophically too large.

This raw failure is informative. It tells us that the observable is not the unfiltered Laplacian of the relic phase field; it must involve additional physical suppression from finite shell thickness, beam smoothing during observation, and a transfer coefficient that is much smaller than unity. Including finite shell thickness , angular smoothing of beam width , and a free transfer coefficient , the filtered observable is:

A sweep of 112 combinations of yields 56 combinations in the target to CMB amplitude range. The best combinations cluster around – (thin-shell approximation holds), – (realistic observational smoothing), and . The peak amplitudes land essentially on the scale, and the angular widths come out in the – range, which is within the sensitivity of modern CMB experiments. The geometric projection survives the thin-shell readout, the observational filtering produces realistic amplitudes and widths, and the required transfer coefficient is consistent across the best combinations.

The compound transfer coefficient. The required value is not mostly a detector artifact. A direct extraction comparing raw and filtered projections shows that the geometric and detector suppression factor (from finite shell thickness and beam smoothing) stays close to unity in the best rows — typically –, dropping to only in the heaviest smoothing cases. Therefore:

The transfer coefficient behaves like a real thermodynamic conversion factor, not a geometric blur. This is the cleanest numerical result of the filtered-projection program: the required suppression is mostly physical and mostly constant across the target parameter space, and it sits at approximately .

What kind of physical quantity could produce a transfer coefficient of ? The natural candidates are simple abundance ratios — the baryonic fraction , the radiation fraction at the relevant epoch, the temperature conversion factor from the Stefan-Boltzmann relation , or products of these. Four hypotheses were scanned:

H1: . Zero hits in the target ballpark.

H2: . Zero hits.

H3: . Few grazing hits, demonstrating that the radiation temperature-conversion factor matters but the result still undershoots.

H4: with additional factor . 50 hits in the target ballpark, with best fits clustered near .

Simple abundance partition is ruled out. The required transfer coefficient is compound: it involves the temperature conversion factor, matter and radiation abundance fractions, and an additional residual efficiency of approximately .

The candidate splice. The value is suggestively close to . Imposing the candidate identification:

and scanning across plausible matter and radiation fractions yields 17 direct ballpark hits, with the best cases landing almost exactly on the target . Example parameter points include giving , giving , and giving . The splice is numerically viable across mild parameter ranges and substantially cleaner than the purely phenomenological fit. It is the fifth appearance of the structural identity that Section 7 introduced at candidate status.

The physical interpretation offered for this splice is that encodes an angular-to-linear phase-space conversion. The internal Compton clock ticks in an angular phase space bounded by per complete cycle. The CMB observable is a linear thermodynamic measurement in a three-dimensional spatial volume that projects onto a two-dimensional last-scattering surface. Projecting an internal angular variance onto an external linear space geometrically requires dividing by the product of the two angular domains, which is . The interpretation is physically motivated and numerically preferred, and it connects the fifth appearance to the same structural mechanism that produces the other four appearances — the geometric interface between angular quantum phase evolution and linear holographic boundary accounting.

The honest epistemic status. The fifth appearance is presented as a candidate splice, not a derived theorem. Specifically: the numerical match across plausible parameter ranges is strong, the physical interpretation as angular-to-linear conversion is motivated and consistent with the rest of the framework, and the splice is substantially cleaner than phenomenological alternatives. But the unbroken symbolic derivation — writing the chain from the relic phase perturbation through the photon fluid response to the observed anisotropy , with the factor attached to an explicit angular-to-linear conversion step — is not yet in hand. Until that chain is written, the fifth appearance remains at candidate-derivation status: stronger than a coincidence, weaker than a theorem. This is Open Theorem 3 in Section 14, and the paper is explicit about the distinction both here and in the Section 7 unification table.

A related open question: whether the angular-to-linear of the fifth appearance is the same factor as the phase-convention boundary-bulk decomposition of Appearances 1 through 4, or a consistent extension of it, or a genuinely different that happens to share the numerical value. The current paper takes no position on this question and flags it explicitly as part of Open Theorem 3. The distinction is not just bookkeeping: if the two factorizations are the same, then the fifth appearance is a natural extension of the existing decomposition and its derivation closure should be straightforward; if they are different, then the framework may contain two distinct factors that coincide numerically, and the structural unification would require deeper explanation.

What this section establishes. The early-universe shatter is a forced thermodynamic transition analogous in structure to the late-universe engine crash, driven by thermal cooling through the matter-radiation handoff rather than by structural exhaustion, and producing a

relic phase perturbation whose filtered 3D-to-2D projection onto the last-scattering surface is the observed CMB temperature anisotropy. The required transfer coefficient is compound, sitting at approximately , and a candidate splice identifies the residual efficiency with the structural factor . This is the fifth appearance of in the framework, presented at honest candidate-derivation status with a physically motivated interpretation as angular-to-linear phase-space conversion and an unbroken symbolic derivation flagged as Open Theorem 3. The early-universe section is deliberately more tentative than the late-universe section, reflecting the actual state of the research: the shatter framework is structurally viable and quantitatively constrained, but the detailed derivation chains are still being closed.

12. The Strong-Field Closure: Exact General Relativity

Sections 5 through 11 developed the cosmological, galactic, and early-universe sectors of Solvency Field Theory, all of which operate in regimes where the local gravitational curvature is small compared to the scale of cosmological information production. This section addresses the opposite regime: compact objects, where the maintenance cost is dominated locally and the screening picture reduces to standard strong-field geometry. The claim of this section is that in the strong-field limit, SFT does not modify general relativity — it recovers the exact Schwarzschild exterior metric and the exact Tolman-Oppenheimer-Volkoff hydrostatic equations as the leading strong-field closure of the local information-maintenance balance. Earlier versions of the framework asserted this recovery without deriving it; this section presents the structural argument that makes the exponential metric form the natural consequence of the local balance, with the full uniqueness proof deferred to the companion.

The local compactness parameter. The fundamental object in the strong-field analysis is the local compactness:

which measures the ratio of gravitational potential energy to rest-mass energy at radius from a mass . For the system is in the weak-field regime (galaxies and diffuse large-scale structure, governed by the Section 6 rotation-curve analysis and its screening-energy extension). For the system approaches the Schwarzschild limit (horizons, black holes, and the densest stellar remnants). The range covers everything from the Earth's surface gravity () to neutron star interiors (), and it is the range within which the local information-maintenance balance must produce a continuous gravitational description.

The local analog of the master equation. In the cosmological sector, the master equation relates the expansion rate of the universe to the ratio of bulk information production and boundary information capacity, both integrated over the Hubble volume. In the strong-field sector, an analogous relation holds locally: at each radius in a spherically symmetric static mass distribution, the local information production rate must equal the local boundary accommodation rate, or the maintenance ledger becomes insolvent at that radius. The local balance condition can be written as:

where is the local rate of Compton maintenance production per unit volume (a function of the local rest-mass density and the local Compton frequency) and is the local rate at which the surrounding geometry can accommodate that production through its curvature. The detailed form of both sides follows from applying the Compton Maintenance Postulate locally and computing the boundary accommodation rate from the local metric. The full derivation of both sides is in the companion document.

The structural argument for the exponential metric form. Within the current strong-field construction, the exponential metric form is the leading candidate singled out by the requirement that the local Landauer cost be absorbable at every radius. The structural argument proceeds as follows. Three-sentence form:

Starting point. The local information-maintenance balance, when applied to a spherically symmetric static metric with arbitrary radial function, produces an algebraic condition relating the metric components and to the local compactness and the rest-mass density . Assume the metric takes the form and for some function , which is the form consistent with spherical symmetry, staticity, and the asymptotically-flat signature convention.

Pivot. Substituting this ansatz into the local balance condition produces a first-order ordinary differential equation for of the form , where is determined by the local compactness. The key structural point is that the equation is first-order in , not in itself — which means that any solution is necessarily an exponential in the integrated source function . Under the asymptotic flatness boundary condition as , the unique solution of this first-order ODE is:

where is the scalar potential determined by integrating the source from infinity inward. Any other functional form would either fail to satisfy the first-order ODE or fail to satisfy the asymptotic boundary condition, and in either case would produce a radial function whose Landauer cost cannot be locally absorbed by the pre-existing metric curvature at some radius — violating the maintenance balance at that point. The companion document carries the detailed derivation of the first-order ODE from the local balance condition and the step-by-step uniqueness analysis; the main paper presents the result as the structural consequence of the first-order structure.

Result. The local balance condition, combined with the spherically symmetric static ansatz and the asymptotic flatness boundary condition, forces the metric to take the form , , where is the scalar potential determined by integrating the local balance source from infinity. This is the structural result that earlier versions of the framework asserted without deriving; the main paper presents it here as the first-order-ODE consequence of the local balance with the full symbolic uniqueness proof deferred to the companion.

Recovery of the Schwarzschild metric. With the exponential form established, the radial function is determined by solving the first-order ODE with the Schwarzschild boundary conditions: as (asymptotic flatness) and determined by the total enclosed mass at each radius. For a point mass in vacuum, the integration yields:

and the resulting metric is:

This is the exponential-coordinate form of the Schwarzschild metric. In the standard isotropic-coordinate representation, the same physical metric is written as:

which is the standard Schwarzschild exterior metric. The two forms are related by a coordinate transformation and are physically identical. The point is that SFT's local information-maintenance balance, with no free parameters and no approximations, selects the Schwarzschild metric as its leading strong-field solution. It is not a post-Newtonian expansion, not a leading-order match to general relativity, and not a modification of the Einstein field equations. It is the exact vacuum exterior metric that any theory of gravity must produce for a spherically symmetric point mass, and SFT produces it through an information-theoretic argument rather than through field equations.

Recovery of the TOV equations. For the interior of a self-gravitating fluid with pressure and density , the local balance condition extends naturally: the pressure gradient required to support a fluid element against gravitational collapse is exactly the gradient that the maintenance balance requires to keep the information budget solvent at every radius. The full derivation proceeds by applying the local balance condition to a fluid element with nonzero pressure, substituting the exponential metric form, and reading off the hydrostatic equilibrium condition. The resulting equation is:

which is the exact Tolman-Oppenheimer-Volkoff hydrostatic equilibrium equation used in standard relativistic astrophysics to compute neutron star and white dwarf structure. The TOV equation is not modified, approximated, or extended by SFT; it is recovered exactly as the interior form of the local information-maintenance balance for a pressurized self-gravitating fluid. Neutron star equation-of-state calculations, stellar interior structure computations, and any other application of the TOV equation continue to work in SFT without modification.

The weak-field limit and the Branch C* connection. At low compactness , the exponential metric form can be expanded:

The leading-order term reproduces the Newtonian limit of general relativity and recovers the standard post-Newtonian expansion. The key rigor check is that the exponential form passes the classical post-Newtonian gates exactly. The Eddington parameter , which measures the deviation of the metric from pure Newtonian gravity in the second-order terms, comes out to exactly — the value required by solar-system tests and by all existing measurements of gravitational lensing and light deflection. The first-order light deflection, computed from the exponential metric, gives the Eddington result of for a photon with impact parameter , again exactly matching the general-relativistic prediction and the observed deflection of starlight during solar eclipses. These weak-field / solar-system gates are *passed*; the full nonlinear field-theoretic extension of the framework, bridging the strong-field closure to the cosmological screening sector, is flagged as Open Theorem 6 in Section 14.

The full analysis of the weak-field lag-action extending the exponential-metric solution into the weak-field source-sector is developed as the Branch C* program, documented in the companion document. The central claim of Branch C* is that the exponential metric form, once justified as selected by the local balance condition rather than postulated, passes all classical post-Newtonian tests and extends consistently into the weak-field source sector where the cosmological screening mechanism of Sections 6 and 10 becomes dynamically relevant. The bridge between the Branch C* source sector and the cosmological screening sector is the subject of ongoing work (Open Theorem 6 in Section 14), but the strong-field and weak-field limits are both handled by the same exponential metric form, and the transition between the two regimes is continuous through the compactness parameter .

Interpretation: SFT provides a foundation for general relativity. The central structural claim of this section is that SFT does not modify, replace, or extend general relativity. In the strong-field regime, SFT produces exactly the Schwarzschild metric and exactly the TOV equations as the leading closure of the local balance condition; these are not approximations. General relativity emerges as the strong-field limit of the information-maintenance principle, just as Newtonian gravity with cosmological screening corrections emerges as the weak-field limit in Section 6. The screening mechanism itself — the thing that makes galactic rotation curves deviate from Newtonian predictions below — is the bridge between the two regimes. At high compactness, the maintenance cost is fully absorbed locally by the pre-existing gravitational curvature, the cosmological contribution to the local balance is negligible, and standard general relativity governs. At low compactness, local screening is insufficient to absorb the full maintenance cost, the cosmological contribution becomes dynamically relevant, and the weak-field deviations from pure Newtonian behavior appear. The screening mechanism is not an add-on to SFT; it is the structural bridge between the two regimes of the same underlying principle.

What this means is that SFT is not a competitor to general relativity. It is an information-theoretic foundation for general relativity, in the same sense that statistical mechanics is a foundation for thermodynamics. Thermodynamics works without reference to statistical mechanics — you can compute heat engines and entropy production using thermodynamic relations alone — but statistical mechanics tells you what thermodynamics is *about*: the aggregate behavior of many microscopic degrees of freedom. In the same way, general relativity works without reference to SFT — you can compute black hole interiors and gravitational lensing using the Einstein field equations alone — but SFT tells you what general relativity is *about*: the geometric response of a universe that pays for its own irreversible computation. The Schwarzschild metric is not an SFT modification of a GR solution; it is the SFT expression of what the GR Schwarzschild solution has been describing all along, from a different starting point.

What this section establishes. Solvency Field Theory's local information-maintenance balance admits exact general-relativistic solutions as its leading strong-field closure. The exponential metric form is selected structurally as the unique solution of a first-order ODE under the asymptotic flatness boundary condition, with the full symbolic uniqueness proof deferred to the companion. The resulting metric is exactly Schwarzschild in vacuum and exactly TOV for interior fluids, and the weak-field / solar-system post-Newtonian gates are passed exactly (, light deflection). The full nonlinear field-theoretic extension of the framework, bridging the strong-field closure to the cosmological screening sector through the

Branch C* lag-action program, is flagged as an open theorem. SFT is not a modification of general relativity; it is an information-theoretic foundation that recovers general relativity's strong-field sector as the forced closure of the same axioms that produce the cosmological and galactic results of the earlier sections.

13. Falsifiable Predictions

Solvency Field Theory is a falsifiable framework in the strong sense. Each of the results developed in Sections 5 through 12 risked a specific numerical prediction derived from the axiomatic foundation with zero free parameters, and each of those predictions could have killed the framework if the data had gone the other way. Several of the risks have already paid off; the remainder are testable with current or forthcoming observational programs. This section summarizes the predictions in a form a referee can evaluate against current data, and it is explicit about which predictions have been confirmed, which are consistent with existing observations pending stronger tests, and which are awaiting data that does not yet exist.

The predictions are grouped by observational regime rather than by section of the paper. The epistemic status of each prediction is stated explicitly.

Galactic scale: the acceleration threshold and its morphology dependence. The framework predicts that galactic rotation curves deviate from Newtonian behavior below the acceleration threshold m/s, and that the interpolation function contains a morphology-dependent coefficient taking values in the range to depending on the local screen geometry. Both predictions are derived from the Compton Maintenance Postulate and the screen-energy projection mechanism with zero adjustable parameters.

Status: confirmed. The threshold prediction has been tested against 137 SPARC galaxies with 63% rotation-curve error reduction relative to baryons-only and with the derived value outperforming nearby alternatives (alone,) in direct comparison. The morphology dependence has been confirmed against the same sample through a partial-correlation test controlling for additional morphological and photometric variables, with the bulge-fraction correlation significant at . A 137-galaxy sample in which the literal scale outperformed the derived , or in which showed no morphological dependence, would have falsified the theory. Neither happened.

Cosmological scale: the dark energy equation of state and phantom crossing. The framework predicts, through the historical cosmology pipeline, a specific evolution of the dark energy equation of state: and in the CPL parameterization, with the phantom crossing occurring at . The framework also predicts that the modifications to the expansion history emerge only at late times, preserving standard early-universe physics at the CMB epoch, with the acoustic peaks of the Planck 2018 temperature power spectrum compatible to within .

Status: historical pipeline confirmed; corrected-chain reconciliation under audit. The historical cosmology pipeline, based on the evolving channel history extracted from TNG300 and mapped through the earlier working formulation, produces the CPL pair at a distance of 0.005 from the Pantheon+ supernova best-fit, consistent with DESI DR2 BAO constraints at 3.4–5.4, with the phantom crossing at matching the DESI-preferred crossing redshift. CMB compatibility has

been tested through the CLASS Boltzmann solver against Planck 2018, producing an RMS residual of across multipoles –, with the acoustic peak region at and the damping tail at . The underlying channel history and the qualitative crossing-redshift result are stable across both the historical pipeline and the dimensionally corrected reconstruction of from Section 10. Full reconciliation of the specific CPL coefficient pair under the corrected dimensionally consistent chain is deferred to a V10.1 audit update. The CMB residual prediction and the phantom crossing redshift are independent of the audit and remain confirmed. A DESI measurement of outside the predicted CPL region, or a CMB residual that exceeded the acoustic-peak tolerance by more than a factor of a few, would have falsified the historical pipeline. Neither happened.

Cosmological scale: environment-dependent structure growth. The screening mechanism predicts that the effective expansion rate — and therefore the rate of gravitational growth — depends on the local gravitational environment. Overdense regions (walls, filaments, halos) have more gravitational curvature to absorb the Compton maintenance cost locally, reducing the effective expansion contribution and enhancing structure growth. Underdense regions (voids) have less local absorption, transmitting more of the maintenance cost to the cosmological boundary and suppressing growth relative to the mean. The predicted quantitative signature is a – growth-rate split between the deepest voids () and modest overdensities () at , measurable through the growth parameter stratified by environment.

Status: testable with DESI DR3 and Euclid. The prediction is sharper than any CDM-based model, which predicts exactly zero environment dependence because a spatially uniform cosmological constant produces identical expansion in all regions. A DESI DR3 or Euclid measurement of environment-stratified that showed zero split, or a split of the wrong sign, would falsify SFT's screening mechanism. A confirmation would be the strongest single piece of evidence for the theory beyond what SPARC and DESI DR2 have already provided.

Cosmological scale: supervoid ISW anomaly. The same screening mechanism predicts that large cosmic voids should produce enhanced integrated Sachs-Wolfe signals in the cosmic microwave background — photons traversing supervoids encounter gravitational potentials that decay faster than a uniform cosmological constant would produce, because the reduced screening inside the void accelerates the local dark-energy-like contribution. The signal is a quantitative excess in the ISW cross-correlation between CMB temperature and supervoid positions.

Status: consistent with existing data. Granett, Neyrinck, and Szapudi reported in 2008 an anomalously strong ISW signal from a sample of supervoids, exceeding CDM expectations by a factor of 2–3. The anomaly has persisted through multiple independent analyses and has no standard explanation within CDM. SFT's screening mechanism provides a natural explanation: supervoids have minimal local screening, so their gravitational potentials decay faster than uniform- predictions, producing exactly the enhanced ISW signal observed. This is consistent with existing data in the sense that the observed anomaly matches the framework's prediction, pending independent reanalysis and cross-checks with forthcoming CMB-S4 data.

Strong-field scale: post-Newtonian tests and compact-object structure. The framework predicts that the strong-field regime is described by exactly the Schwarzschild metric and exactly the Tolman-Oppenheimer-Volkoff hydrostatic equations, through the structural argument of Section 12. This implies specific values for the post-Newtonian parameters: the

Eddington parameter exactly, and the first-order light deflection of for a photon with impact parameter .

Status: weak-field gates confirmed; strong-field exact recovery claimed structurally. The Eddington parameter has been measured to better than one part in through radar ranging of the Cassini spacecraft and lunar laser ranging, with the measured value consistent with at the current sensitivity. Light deflection has been measured in solar eclipse experiments since 1919 and more recently through precision radio interferometry, with the measured deflection consistent with the Eddington value at the current sensitivity. The strong-field exact recovery of Schwarzschild and TOV — while structurally forced by the local balance condition as derived in Section 12 — is supported by all existing observations of neutron star structure and black hole spacetimes, which continue to produce results consistent with exact general relativity at the current level of observational sensitivity. A measurement of at high significance, or a confirmed neutron star structure inconsistent with exact TOV solutions at the current level of equation-of-state uncertainty, would falsify the strong-field closure of the theory. Neither has happened.

A note on the candidate CMB transfer coefficient. Section 11 presented a candidate splice for the CMB transfer coefficient, in which the fifth appearance of enters as . This is a *candidate-derivation status* result, not a confirmed prediction. The framework is explicit about the distinction: the numerical match is strong across plausible parameter ranges and the physical interpretation is motivated, but the unbroken symbolic derivation from the relic phase perturbation to the observed temperature anisotropy is not yet in hand (Open Theorem 3). The candidate splice is not listed among the confirmed predictions above because its epistemic status does not yet meet the threshold for inclusion. It is listed here as a research target rather than a prediction: if the derivation is closed, the candidate splice becomes a new falsifiable prediction about the detailed structure of the CMB transfer function that would be testable against forthcoming spectral-distortion measurements from CMB-S4.

Summary. Of the five load-bearing predictions grouped above, three are confirmed against current data with zero free parameters (galactic acceleration threshold, weak-field post-Newtonian gates, CMB acoustic peak preservation), one is confirmed in its historical-pipeline form with CPL-coefficient reconciliation under audit (dark energy equation of state through the corrected chain), one is consistent with existing data pending stronger tests (supervoid ISW anomaly), and one is awaiting direct measurement from forthcoming surveys (environment-dependent structure growth). The fifth appearance of in the CMB transfer coefficient is a candidate-derivation result, not yet a prediction, and will become a prediction if and when the derivation is closed.

The framework is falsifiable in the strong sense because each of the confirmed predictions could have failed: a SPARC sample in which outperformed would have killed the theory, a DESI measurement of inconsistent with the predicted CPL region would have killed the historical pipeline, a CMB residual exceeding the preserved-peak tolerance would have killed it, and measurements of or TOV-inconsistent neutron star structure would have killed the strong-field closure. None of these failure modes materialized. The forthcoming tests — environment-dependent growth with DESI DR3 and Euclid, supervoid ISW cross-correlation with CMB-S4, morphology-dependent with larger galaxy samples from SKA and WALLABY — each provide another opportunity to kill the theory or to strengthen it substantially.

14. Open Theorems and Research Targets

Solvency Field Theory is a research program, not a finished theory. The framework has produced confirmed predictions across multiple regimes and a structural unification that links the cosmological, galactic, strong-field, and early-universe sectors, but several open questions remain — some of them specific theorems that the framework should be able to prove from its foundation, and others of them empirical validation tasks that require data or cross-checks not yet available. This section separates the two categories explicitly. Subsection 14.1 lists the genuine open theorems — specific mathematical statements the framework is working toward proving. Subsection 14.2 lists the research targets — empirical and methodological tasks whose resolution does not require new theoretical development but does require data or computational resources that the framework has not yet accessed.

14.1 Open theorems

Theorem 1: The formal unification of the shatter and the engine crash. Section 9 claimed structural unification of the early-universe and late-universe forced thermodynamic transitions at the level of physical principle and mathematical scaffolding. The formal unification — a single closed equation whose two solutions are the shatter at matter-radiation handoff and the engine crash at baryonic fuel exhaustion — is not yet in hand. The candidate approach is to generalize the maintenance budget to a function of both temperature and structural fraction , such that the critical condition has two distinct solutions at opposite ends of cosmic history, and to show that the generalized budget reduces to the shatter equations at high and low , and to the engine-crash equations at low and high . Closing this theorem would upgrade the structural unification of Section 9 from a parallel claim about two instances to a formal claim about one equation.

Theorem 2: The second from first principles. Section 7 identified the structural identity as decomposing into two independent factors of — one from the phase convention (versus) and one from the boundary-versus-bulk normalization (Bekenstein-Hawking versus Einstein). The first factor is derived from the Compton Maintenance Postulate directly. The second factor is identified algebraically as the ratio between the two normalization conventions but is not yet derived from first principles. Closing this theorem would show that apparent-horizon thermodynamics requires the specific Bekenstein-Hawking normalization and that the Einstein coupling follows as the corresponding bulk expression, with the ratio forced rather than conventional. The candidate approach is to work directly from the apparent-horizon thermodynamics program of Jacobson and Padmanabhan, deriving both normalization conventions from a common starting point and showing that their ratio is exactly.

Theorem 3: The CMB transfer derivation closure. Section 11 presented the candidate splice for the cosmic microwave background transfer coefficient, with the factor identified provisionally as an angular-to-linear phase-space conversion. The unbroken symbolic chain from the relic phase perturbation through the photon fluid response to the observed temperature anisotropy , with the factor attached to an explicit angular-to-linear conversion step, is not yet written. Closing this theorem would promote the fifth appearance of from candidate-derivation status to derived status and would complete the structural unification of Section 7. A sub-question: determining whether the angular-to-linear of this closure is the

same factor as the phase-convention boundary-bulk decomposition of the other four appearances, a consistent extension of that decomposition, or a genuinely independent that happens to share the numerical value. The resolution of this sub-question has implications for whether the framework contains one fundamental geometric factor or two distinct factors that coincide numerically.

Theorem 4: The morphology-to- derivation from first principles. Section 6 established that the morphology-dependent coefficient in the galactic interpolation function takes the benchmark values (derived spherical limit) and (strong disk-sector candidate), both emerging from the spherical-screen projection integrals applied to different screen geometries. The SPARC data confirm a morphology dependence at . What remains open is the derivation of the mapping from galaxy morphology to screen geometry: given a galaxy's observed mass distribution (disk scale length, bulge fraction, central density profile), which screen geometry applies, and how does the transition between pure disk and pure bulge limits occur for galaxies with intermediate morphologies? The candidate approach is to derive the screen geometry from a variational principle that minimizes the local information-maintenance cost for a given baryonic mass distribution, which would produce a continuous family of screens parameterized by the galaxy's morphological variables.

Theorem 5: The structural unification of the projection integrals. Section 6 produced the benchmark values at galactic scales; Section 10 produced and at cosmological scales. Both pairs emerge from spherical-screen second-moment integrals applied to different geometries. The structural parallel is stated in Sections 6 and 10 but has not been upgraded to a formal theorem. Closing this theorem would produce a single projection-integral theorem that outputs all four values — , , , — from a single underlying geometric construction applied at different scales, with the distinctions between disk, bulge, wall, and filament emerging as limits of a common integral rather than as independent calculations. The candidate approach is to formulate the projection integral on a family of screen geometries parameterized by dimensionality (1D filament, 2D wall/disk, 3D bulge) and angular anisotropy, and to show that the four numerical values emerge as the four limits of the common parameter family.

Theorem 6: The extension beyond FLRW minisuperspace. The constrained action of Section 8 is formulated on the FLRW minisuperspace, describing only homogeneous isotropic cosmologies. The full field-theoretic extension, addressing perturbation theory beyond background dynamics and allowing inhomogeneous and anisotropic configurations, remains open. The weak-field lag-action program (Branch C*, mentioned in Section 12) is a first step in this direction, handling the post-Newtonian perturbations around a Schwarzschild background. The full extension would produce a complete field-theoretic SFT formulation whose background reduction is the constrained minisuperspace action and whose perturbation sector reproduces standard cosmological perturbation theory at leading order while introducing the SFT-specific corrections from the maintenance screening mechanism. This is the largest single theoretical task remaining for the framework.

14.2 Research targets

Target 1: CPL coefficient reconciliation under the corrected chain. Section 10 presents both the historical cosmology pipeline (which produced , with phantom crossing at) and the dimensionally corrected derivation from . A first coarse reconciliation rerun preserves the

crossing scale () but shifts the fitted CPL coefficients substantially. The resolution of this target requires running the corrected reconstruction against the full original ten-snapshot TNG300 channel history (or a denser sampling) with smoothed-derivative techniques appropriate to the corrected formula's sensitivity to . This is the highest-priority audit item for V10.1 and does not require new theoretical development — it is a methodological rerun of the existing channel-to-CPL mapping through the mathematically corrected chain.

Target 2: Cross-simulation validation of the cosmological sector. The four-state cosmic web ladder and the derived in Section 10 were extracted from the IllustrisTNG TNG300 simulation. Independent validation against cosmological simulations with different sub-grid physics — EAGLE, SIMBA, Magneticum, and CAMELS-suite variations — would confirm that the ladder and the derived dark energy equation of state are physical results rather than artifacts of the TNG300-specific baryonic feedback prescription. The resolution of this target requires computational resources to process alternative simulation suites and extract the corresponding active-and-passive-channel histories; no new theoretical development is needed.

Target 3: The 13% residual in . The derived galactic acceleration threshold m/s is 13% below Milgrom's empirical m/s. Candidate explanations for the residual include the QCD binding energy contribution to the effective Compton frequency (protons are approximately 99% binding energy rather than rest mass of constituent quarks, which would modify for baryonic matter by a small factor), cosmological evolution of between the epoch when was empirically calibrated and , and systematic uncertainties in the empirical determination of from galaxy samples with varying morphology and data quality. Resolving this residual requires either a first-principles calculation of the QCD binding correction to the effective Compton rate, or a reanalysis of the empirical determination with morphology stratification matching SFT's predicted values.

Target 4: Growth-rate measurement with DESI DR3 and Euclid. The environment-dependent structure growth prediction of Section 13 is testable with forthcoming data from DESI DR3 and the Euclid mission. The resolution of this target requires waiting for the data; no theoretical development is needed. A confirmation would be the strongest single piece of empirical evidence for the framework beyond what SPARC and DESI DR2 have already provided.

What this section establishes

Solvency Field Theory has identified six specific open theorems — statements the framework should be able to prove from its axiomatic foundation but has not yet closed — and four research targets whose resolution depends on empirical data, computational resources, or methodological reruns rather than new theoretical development. The theorems form the framework's current theoretical frontier; the targets form its current empirical and methodological frontier. Closing any of the theorems would strengthen the framework's internal coherence; completing any of the targets would strengthen its empirical anchoring. The framework's credibility as a research program depends on its ability to make progress on both fronts, and the specific theorems and targets listed above are the places where that progress should be measured.

15. Conclusion

The universe has a history because information processing is irreversible; expansion is the shape of that history at cosmological scale, and gravity is its shape at every other scale.

This paper has developed Solvency Field Theory as an information-theoretic foundation for cosmology and gravity. Four axioms drawn from established physics — Landauer's principle (information has thermodynamic cost), the Connes-Rovelli thermal time hypothesis (time is the accumulation of irreversible state changes), the holographic bound (space has finite information capacity), and the identification of cosmic expansion as the geometric cost of maintenance (the universe pays for its own computation through geometry) — jointly ground the master equation as a thermodynamic identity rather than a postulated dynamical law. The Compton Maintenance Postulate provides the physical mechanism: every massive particle maintains its quantum existence through continuous phase evolution at the Compton frequency , each complete phase cycle constitutes one irreversible bit of information, and each bit demands one Planck area of cosmological boundary surface to accommodate it.

The load-bearing results that follow from this foundation, derived with zero free parameters and zero adjustable constants, are:

The modified Friedmann equation , differing from the standard result by a content-independent geometric factor of exactly , derived both algebraically from the master equation and variationally from the holographic count equation with a constrained minisuperspace action.

The galactic acceleration threshold m/s, confirmed against 137 SPARC galaxies at 63% rotation-curve error reduction, outperforming all nearby alternative scales, with morphology-dependent interpolation coefficients (benchmark values and) and a morphology-correlation confirmed against the same sample at .

The structural identity appearing in five distinct physical contexts — the Friedmann ratio, the galactic acceleration scale, the holographic capacity gap, the dark energy structural gap, and the candidate CMB transfer coefficient — in each case decomposing into the same two independent factors of (phase convention boundary-bulk normalization), with four appearances derived and the fifth at candidate-derivation status.

The four-state cosmic web ladder with environmental weights , , , , presented as a structural result from axis factorization and the identity, with static projection corrections (,) derived from the same spherical-screen integrals that produced the galactic morphology parameters.

The dimensionally corrected derivation from the master equation, yielding , which makes visible in the math the mechanism that Section 9 framed structurally: the equation of state is sensitive to how the maintenance regime is evolving, and forced transitions are exactly the moments when that evolution becomes observable.

The historical dark energy equation of state , , reproducing the DESI DR2 measurement at 0.005 in the CPL plane through the established channel-history pipeline, with phantom crossing at matching the DESI-preferred redshift, and CMB compatibility at RMS residual against Planck 2018 across multipoles –. Full reconciliation of the CPL coefficients under the

dimensionally corrected chain is flagged as a V10.1 audit item; the qualitative crossing-redshift result is stable across both chains.

The structural unification of the early-universe and late-universe forced thermodynamic transitions — the shatter at matter-radiation handoff and the engine crash at baryonic fuel exhaustion — as two instances of a single physical principle: the maintenance process undergoes forced transitions when its current regime can no longer sustain the thermodynamic balance, producing observable scars on the cosmic record at both ends of cosmic history.

Exact recovery of the Schwarzschild exterior metric and the Tolman-Oppenheimer-Volkoff hydrostatic equations in the strong-field limit, through a first-order-ODE structural argument showing that the exponential metric form is the solution of the local information-maintenance balance under the asymptotic flatness boundary condition, with the classical post-Newtonian gates (, first-order light deflection) passed exactly.

These results do not require new fields, new particles, new dimensions, or modifications to general relativity. They require only that existence is computational, that time is the accumulation of irreversible computation, and that the universe measures computation in complete phase cycles at the Compton frequency. The dark energy problem, the MOND acceleration scale, and the holographic capacity gap — three puzzles conventionally treated as unrelated and individually unsolved — are not three separate mysteries. They are three measurements of the same geometric identity, viewed from different regimes of the same underlying information-maintenance mechanism.

Solvency Field Theory is falsifiable in the strong sense. Each of the load-bearing predictions risked a specific numerical outcome that could have killed the framework if the data had gone the other way. A 137-galaxy SPARC sample in which the literal scale outperformed the derived would have killed it. A DESI DR2 measurement of outside the predicted CPL region would have killed the historical pipeline. A CMB residual exceeding the preserved-peak tolerance at Planck 2018's sensitivity would have killed it. A neutron-star structure inconsistent with exact TOV solutions would have killed it. None of these failure modes materialized. The forthcoming tests — environment-dependent structure growth with DESI DR3 and Euclid, the supervoid ISW cross-correlation with CMB-S4, the morphology-dependent prediction with SKA and WALLABY rotation curves — each provide another opportunity to kill the theory or to strengthen it substantially.

The framework is not finished. Six open theorems — the formal unification of the shatter and the engine crash, the first-principles derivation of the second , the closure of the CMB transfer derivation chain, the morphology-to- mapping, the structural unification of the projection integrals, and the field-theoretic extension beyond FLRW minisuperspace — constitute the theoretical frontier. Four research targets — CPL coefficient reconciliation under the corrected chain, cross-simulation validation of the cosmological sector, resolution of the 13% residual in , and the DESI DR3 / Euclid growth-rate measurement — constitute the empirical and methodological frontier. The framework's value as a research program is measured by its ability to make progress on both fronts, and the specific theorems and targets listed in Section 14 are the places where progress should be measured.

The thesis of Solvency Field Theory, stated one final time: the universe has a history because information processing is irreversible. Expansion is the shape of that history at cosmological scale. Gravity is its shape at every other scale. The four axioms ground the master equation; the Compton Maintenance Postulate gives it physical content; the structural identity unifies five distinct physical contexts; the forced thermodynamic transitions of the maintenance process link the early universe to the late universe through a single mechanism; the strong-field limit recovers general relativity exactly. These are not separate results. They are one result, viewed from seven different scales of the same underlying principle, and the framework stands or falls on whether that principle describes the universe we actually live in.

The results presented here are consistent with the universe we have observed so far. The tests that will resolve whether this consistency is structural or coincidental are specific, they are achievable with current and forthcoming observational programs, and they are listed explicitly. Solvency Field Theory is the claim that the universe pays for its own computation through geometry, and that every cosmological, galactic, and gravitational observable is a measurement of that payment. Whether the claim is correct is, now, an empirical question.

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